

4DVarNet-FM: Training Experiments (12 runs, T*)

Experiment Hyperparameters

ID	Model	Channels	S1Ep	S2Ep	TimeEmb	N_outer	SigPrior	TrainMix	Rand	ParamNoise
T1_direct_unet_s0s1	DirectUNet	[64, 128, 256]	200	0	-	-	-	s0_s1	no	-
T2_direct_unet_s0s1_small	DirectUNet	[32, 64, 128]	200	0	-	-	-	s0_s1	no	-
T3_joint_direct_unet_s0s1	JointDirectUNet	[64, 128, 256]	200	0	-	-	-	s0_s1	no	-
T4_joint_direct_unet_s0s1_small	JointDirectUNet	[32, 64, 128]	200	0	-	-	-	s0_s1	no	-
T5_vanilla_cfm_s0s1	VanillaCFM	[64, 128, 256]	800	0	64	10	0.5	s0_s1	no	-
T6_vanilla_cfm_s0s1_tau0	VanillaCFM	[64, 128, 256]	400	0	64	10	0.5	s0_s1	no	-
T7_vanilla_cfm_s0s1_small	VanillaCFM	[32, 64, 128]	800	0	64	10	0.5	s0_s1	no	-
T8_vanilla_cfm_s0s1_small_tau0	VanillaCFM	[32, 64, 128]	400	0	64	10	0.5	s0_s1	no	-
T9_joint_cfm_s0s1	JointCFM	[64, 128, 256]	800	0	64	10	0.5	s0_s1	no	-
T10_joint_cfm_s0s1_tau0	JointCFM	[64, 128, 256]	400	0	64	10	0.5	s0_s1	no	-
T11_joint_cfm_s0s1_small	JointCFM	[32, 64, 128]	800	0	64	10	0.5	s0_s1	no	-
T12_joint_cfm_s0s1_small_tau0	JointCFM	[32, 64, 128]	400	0	64	10	0.5	s0_s1	no	-

Shared settings (config/lorenz63_default.yaml):

dt=0.01, T_max=3.0s, obs_interval=20, R_var=0.5, B_var=2.0
num_windows=2000, spinup_steps=10000, batch_size=32, dropout=0.1
optimizer=Adam, gradient_weight=0.1 (grad-loss term)
Stage 2 only ever runs for Tweedie (S2Ep>0 above); DirectUNet/VanillaCFM are stage-1-only.

T* (data_setup=s0_s1): train/val use randomized param_bias in [0, bias_max] and forcing_state_bias=0 (RandomBiasLorenz63Dataset); eval on two held-out test sets:
S0: param_bias=0.0, forcing_state_bias=0.0 (in-distribution, no bias)
S1: param_bias=0.15, forcing_state_bias=0.1, fixed (out-of-distribution bias)

DirectUNet: single UNet1D pass, obs -> state (MSE + gradient loss), no flow matching.
VanillaCFM: conditional flow matching, LinearInterpolant(nu=1.0), Euler sampling over N_outer=10 steps, $x_0 \sim N(0, \sigma_{prior})$.
JointCFM: VanillaCFM backbone jointly conditioned on/predicting model parameters (param_dim, param_loss_weight); train tau_0 only restricts training to tau=0.
JointDirectUNet: DirectUNet backbone with the same joint param-conditioning head (param_dim, param_loss_weight) as JointCFM; no tau (single pass).
Tweedie: 2-stage solver -- stage1 fits a Gaussian mean estimator, stage2 fits a non-Gaussian residual correction on top of the frozen stage1 estimator.

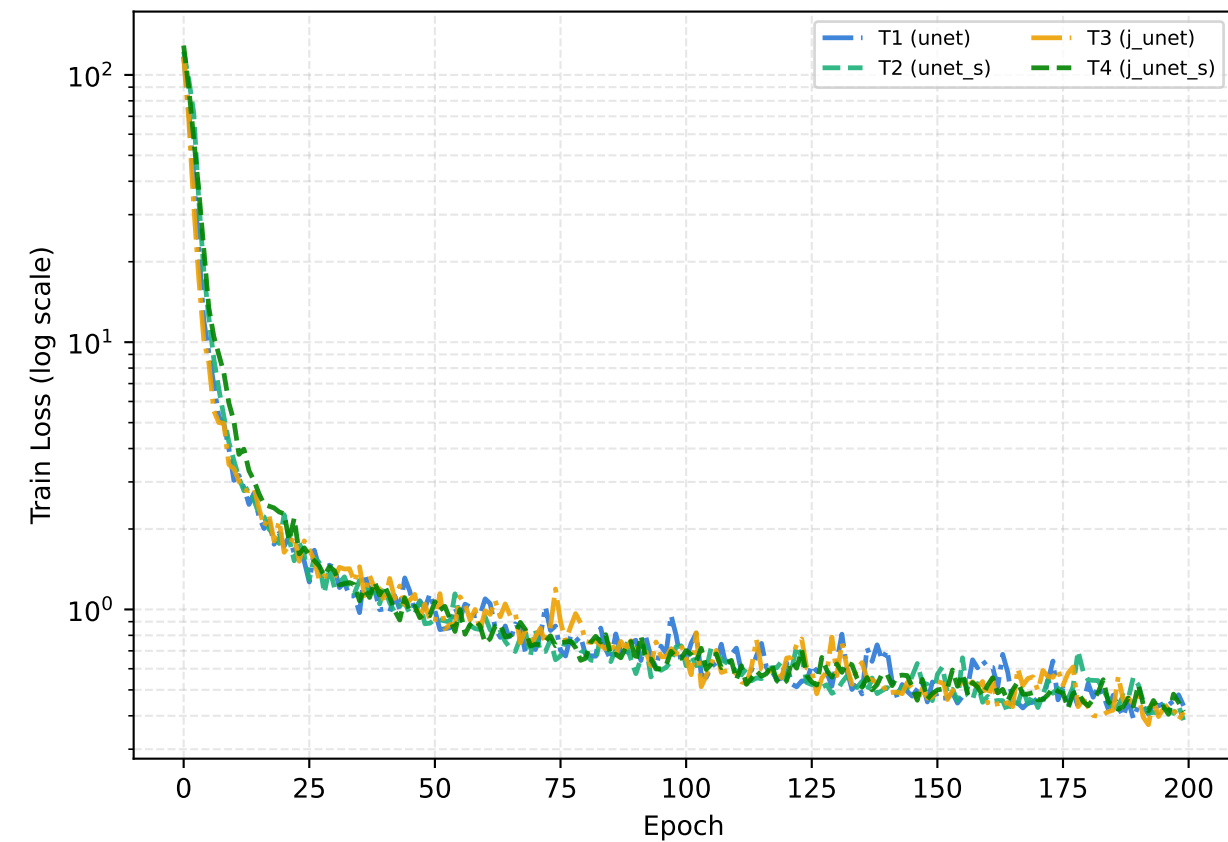
Run Status

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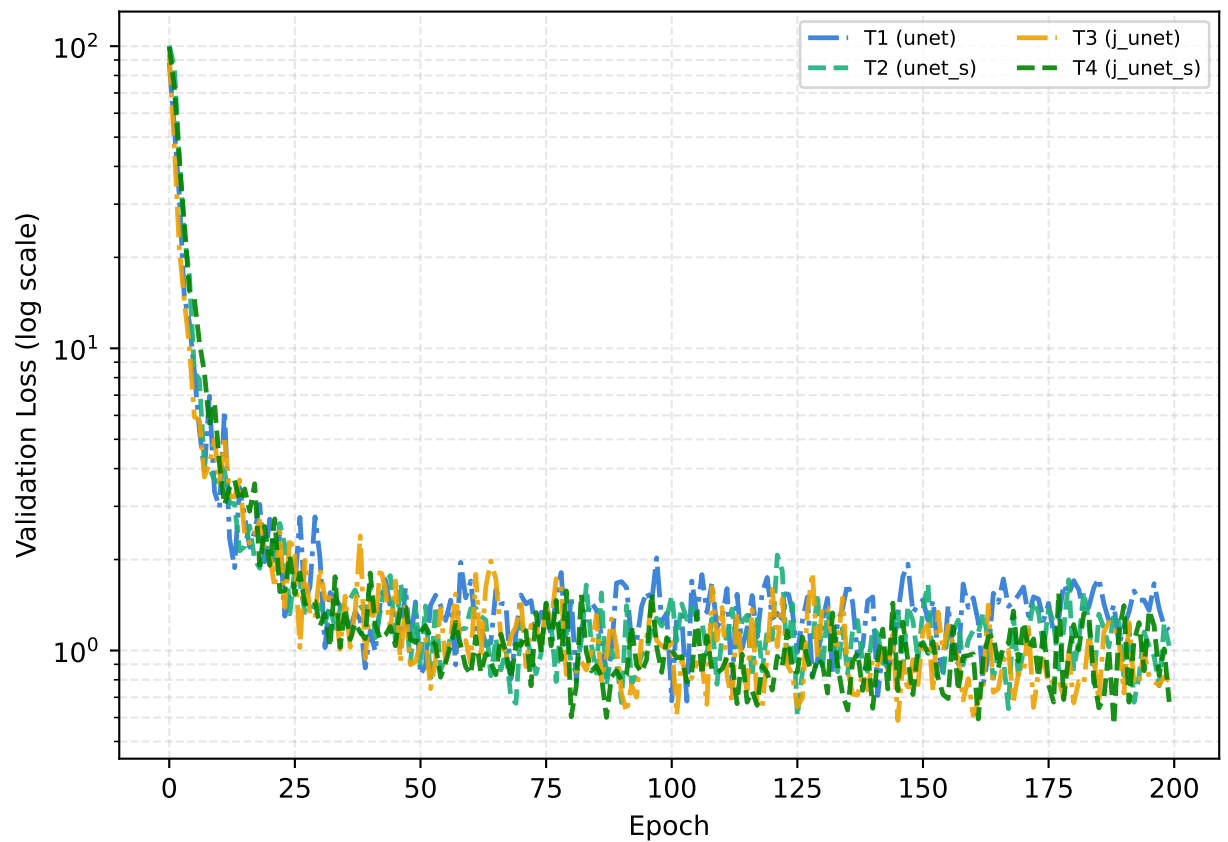
T1_direct_unet_s0s1          trained (results + loss log)
T2_direct_unet_s0s1_small    trained (results + loss log)
T3_joint_direct_unet_s0s1    trained (results + loss log)
T4_joint_direct_unet_s0s1_small trained (results + loss log)
T5_vanilla_cfm_s0s1          trained (results + loss log)
T6_vanilla_cfm_s0s1_tau0     trained (results + loss log)
T7_vanilla_cfm_s0s1_small    trained (results + loss log)
T8_vanilla_cfm_s0s1_small_tau0 trained (results + loss log)
T9_joint_cfm_s0s1           trained (results + loss log)
T10_joint_cfm_s0s1_tau0      trained (results + loss log)
T11_joint_cfm_s0s1_small     trained (results + loss log)
T12_joint_cfm_s0s1_small_tau0 trained (results + loss log)

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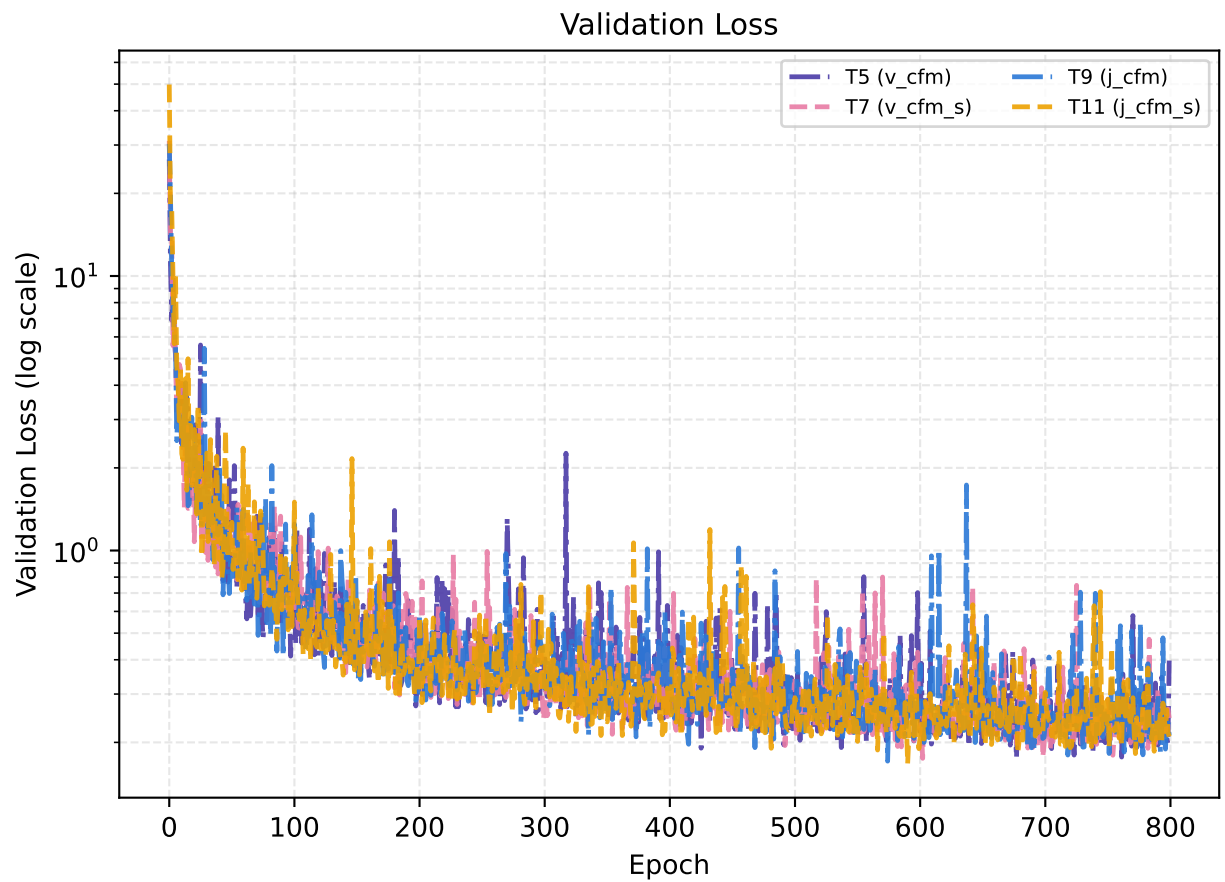
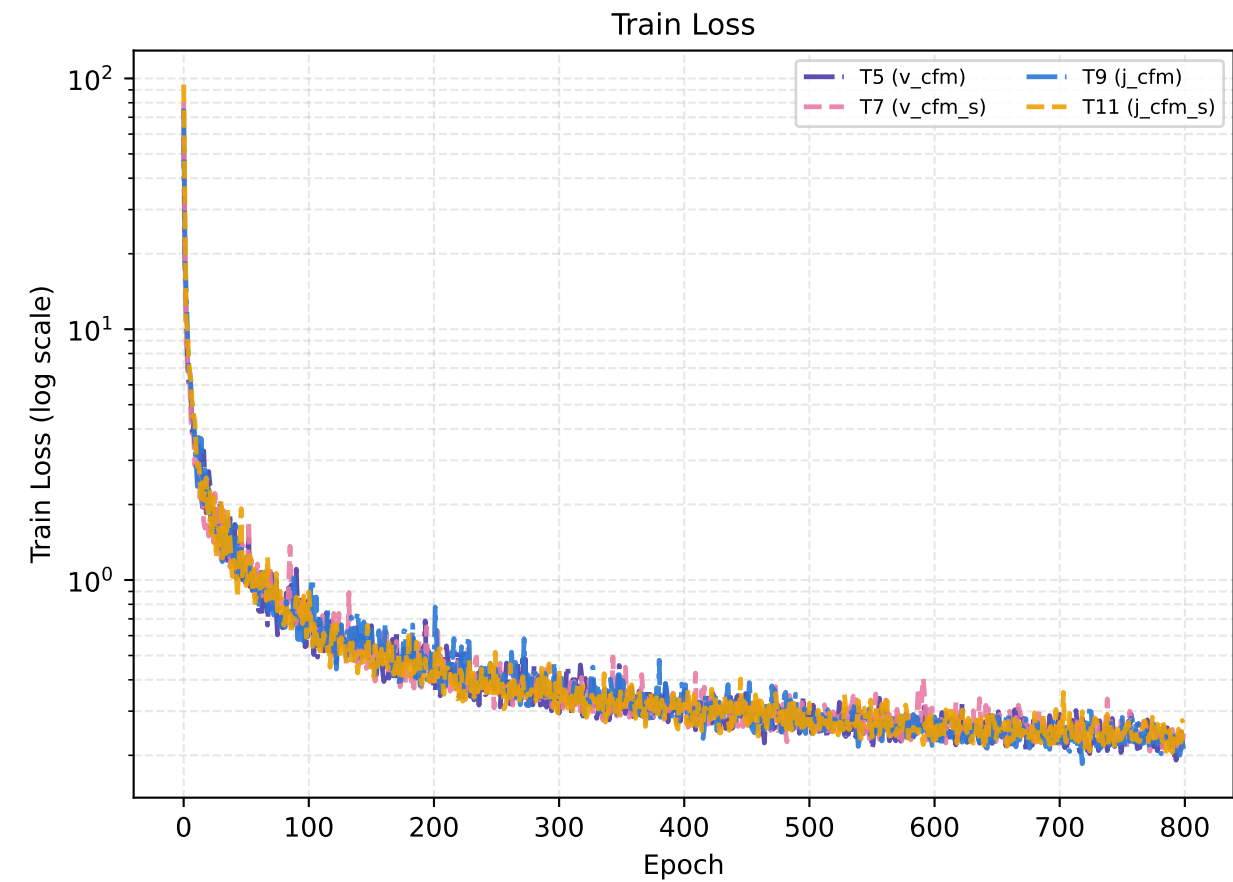
Train Loss



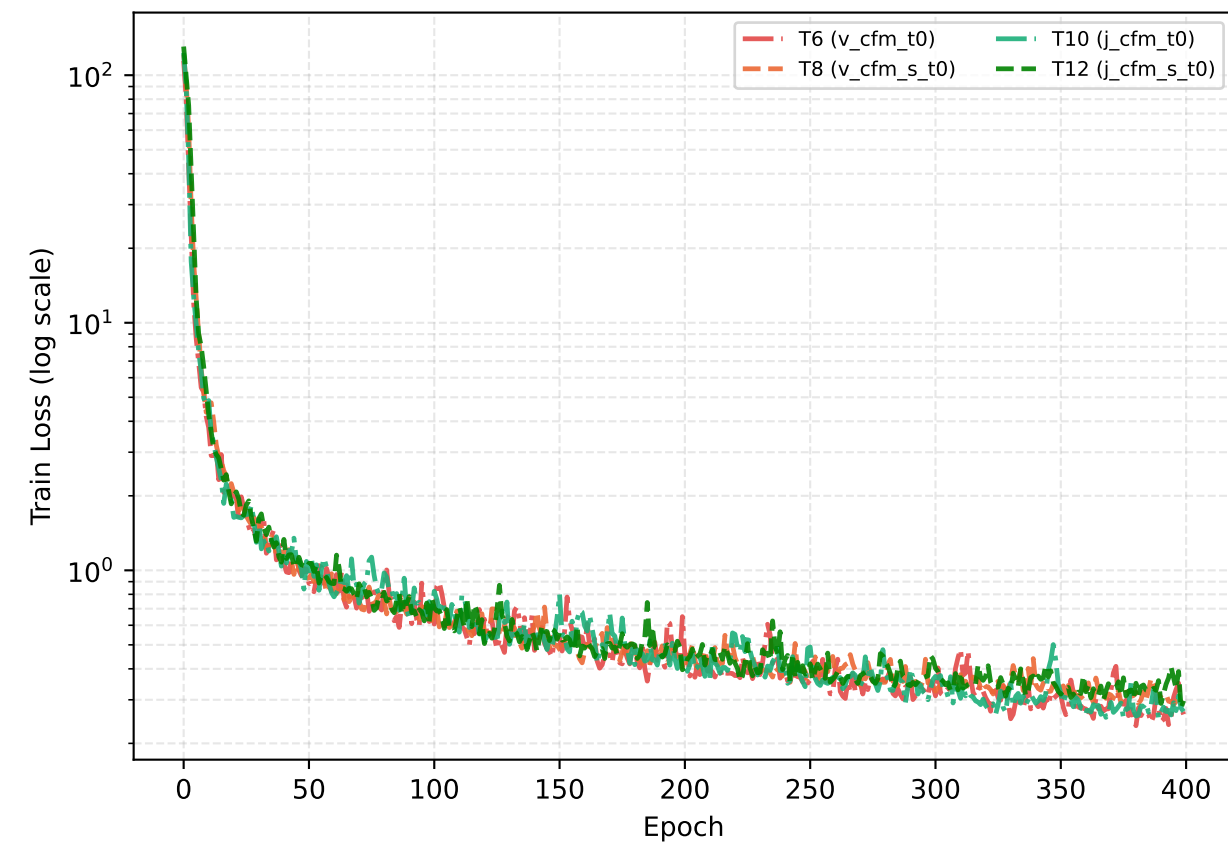
Validation Loss



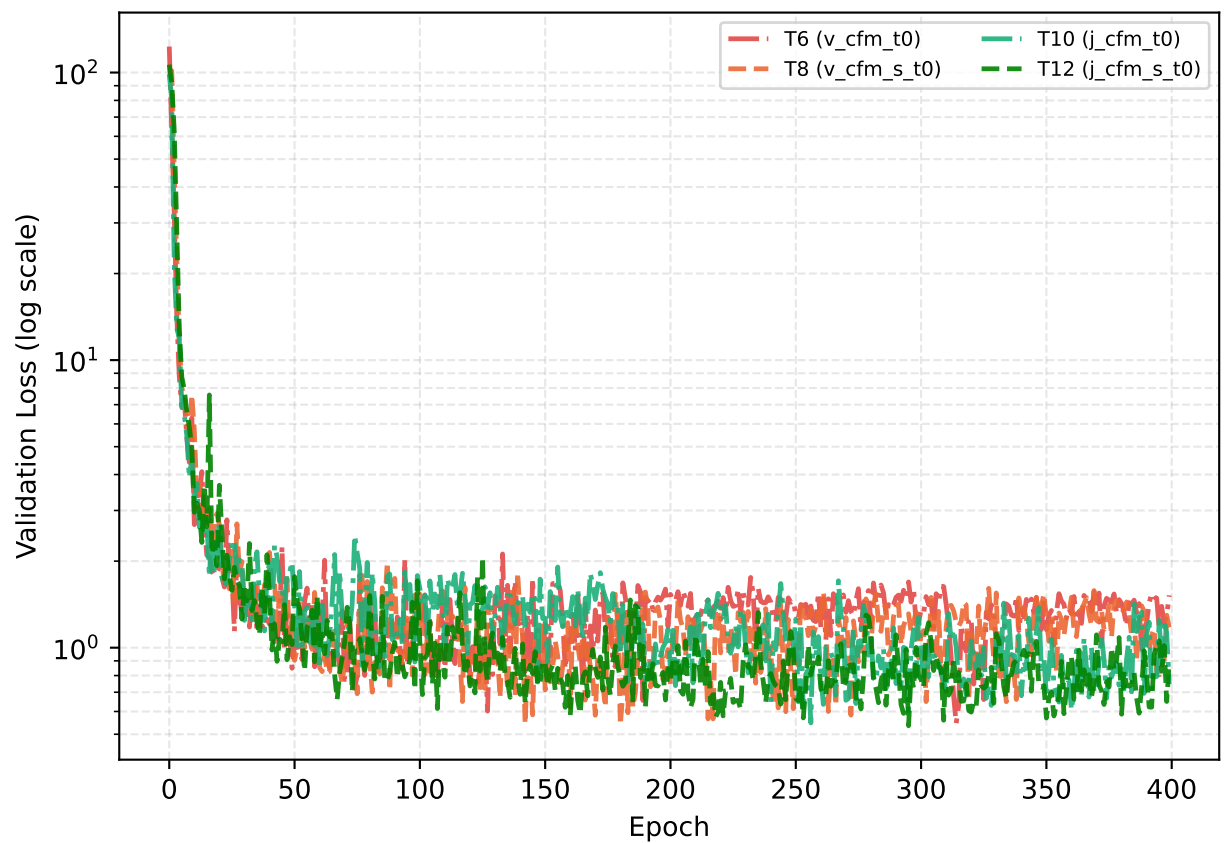
Training & Validation Loss — T5_vanilla_cfm_s0s1 / T7_vanilla_cfm_s0s1_small / T9_joint_cfm_s0s1 / T11_joint_cfm_s0s1_sm



Train Loss



Validation Loss



Per-Variable RMSE Summary (S0 / S1) – X/Y/Z as mean ± std, Mean as plain mean

--- S0 ---

ID	Model	Trained On	X	Y	Z	Mean
T1_direct_unet_s0s1	DirectUNet	s0_s1	0.3609±0.0861	0.5976±0.1304	0.7234±0.1370	0.5606
T2_direct_unet_s0s1_small	DirectUNet	s0_s1	0.3650±0.0746	0.4936±0.1150	0.7743±0.1408	0.5443
T3_joint_direct_unet_s0s1	JointDirectUNet	s0_s1	0.4218±0.0825	0.5490±0.1252	0.7752±0.1490	0.5820
T4_joint_direct_unet_s0s1_small	JointDirectUNet	s0_s1	0.3808±0.0819	0.5133±0.1147	0.6681±0.1132	0.5208
T5_vanilla_cfm_s0s1	VanillaCFM	s0_s1	0.5118±0.1317	0.7372±0.1867	0.8464±0.1273	0.6985
T6_vanilla_cfm_s0s1_tau0	VanillaCFM	s0_s1	0.3967±0.1096	0.5427±0.1815	0.6940±0.1261	0.5444
T7_vanilla_cfm_s0s1_small	VanillaCFM	s0_s1	0.4472±0.0977	0.6172±0.1416	0.7900±0.1341	0.6181
T8_vanilla_cfm_s0s1_small_tau0	VanillaCFM	s0_s1	0.3551±0.0974	0.5002±0.1540	0.6522±0.1281	0.5025
T9_joint_cfm_s0s1	JointCFM	s0_s1	0.4159±0.0930	0.6150±0.1446	0.8667±0.1319	0.6325
T10_joint_cfm_s0s1_tau0	JointCFM	s0_s1	0.3615±0.1001	0.5555±0.1606	0.7107±0.1358	0.5426
T11_joint_cfm_s0s1_small	JointCFM	s0_s1	0.4057±0.0825	0.6172±0.1268	0.8320±0.1457	0.6183
T12_joint_cfm_s0s1_small_tau0	JointCFM	s0_s1	0.3382±0.0869	0.5536±0.1445	0.7304±0.1326	0.5407

--- S1 ---

ID	Model	Trained On	X	Y	Z	Mean
T1_direct_unet_s0s1	DirectUNet	s0_s1	0.7387±0.1193	1.0220±0.2335	2.5152±0.4405	1.4253
T2_direct_unet_s0s1_small	DirectUNet	s0_s1	0.6060±0.1013	0.9186±0.1963	2.6958±0.4290	1.4068
T3_joint_direct_unet_s0s1	JointDirectUNet	s0_s1	0.6410±0.1027	0.9519±0.2109	2.1243±0.3238	1.2391
T4_joint_direct_unet_s0s1_small	JointDirectUNet	s0_s1	0.7147±0.1539	0.9691±0.2449	2.5676±0.4010	1.4171
T5_vanilla_cfm_s0s1	VanillaCFM	s0_s1	0.6421±0.1323	0.9727±0.2272	2.9241±0.4049	1.5130
T6_vanilla_cfm_s0s1_tau0	VanillaCFM	s0_s1	0.7286±0.1380	0.9142±0.1961	2.5735±0.4192	1.4054
T7_vanilla_cfm_s0s1_small	VanillaCFM	s0_s1	0.7213±0.1369	0.9214±0.2457	2.5468±0.3846	1.3965
T8_vanilla_cfm_s0s1_small_tau0	VanillaCFM	s0_s1	0.6399±0.1167	0.8352±0.2043	2.5445±0.4001	1.3399
T9_joint_cfm_s0s1	JointCFM	s0_s1	0.6865±0.1214	0.8231±0.1837	2.3126±0.3977	1.2741
T10_joint_cfm_s0s1_tau0	JointCFM	s0_s1	0.6337±0.1125	0.9132±0.1990	2.4308±0.3704	1.3259
T11_joint_cfm_s0s1_small	JointCFM	s0_s1	0.7322±0.1278	1.0337±0.2165	2.5937±0.4261	1.4532
T12_joint_cfm_s0s1_small_tau0	JointCFM	s0_s1	0.6773±0.1201	0.9588±0.2134	2.7438±0.4390	1.4600

Degradation (S1 mean / S0 mean) and training time:

ID	Model	Trained On	S0 mu	S1 mu	Deg	Time(s)
T1_direct_unet_s0s1	DirectUNet	s0_s1	0.5606	1.4253	2.54	x 772
T2_direct_unet_s0s1_small	DirectUNet	s0_s1	0.5443	1.4068	2.58	x 602
T3_joint_direct_unet_s0s1	JointDirectUNet	s0_s1	0.5820	1.2391	2.13	x 557
T4_joint_direct_unet_s0s1_small	JointDirectUNet	s0_s1	0.5208	1.4171	2.72	x 470
T5_vanilla_cfm_s0s1	VanillaCFM	s0_s1	0.6985	1.5130	2.17	x 2796
T6_vanilla_cfm_s0s1_tau0	VanillaCFM	s0_s1	0.5444	1.4054	2.58	x 1334
T7_vanilla_cfm_s0s1_small	VanillaCFM	s0_s1	0.6181	1.3965	2.26	x 2472
T8_vanilla_cfm_s0s1_small_tau0	VanillaCFM	s0_s1	0.5025	1.3399	2.67	x 1228
T9_joint_cfm_s0s1	JointCFM	s0_s1	0.6325	1.2741	2.01	x 3120
T10_joint_cfm_s0s1_tau0	JointCFM	s0_s1	0.5426	1.3259	2.44	x 1350
T11_joint_cfm_s0s1_small	JointCFM	s0_s1	0.6183	1.4532	2.35	x 2486
T12_joint_cfm_s0s1_small_tau0	JointCFM	s0_s1	0.5407	1.4600	2.70	x 1270

Parameter RMSE Summary (S0 / S1) — Joint Variants

Parameter RMSE Summary (S0 / S1) — Joint Variants

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--- S0 ---

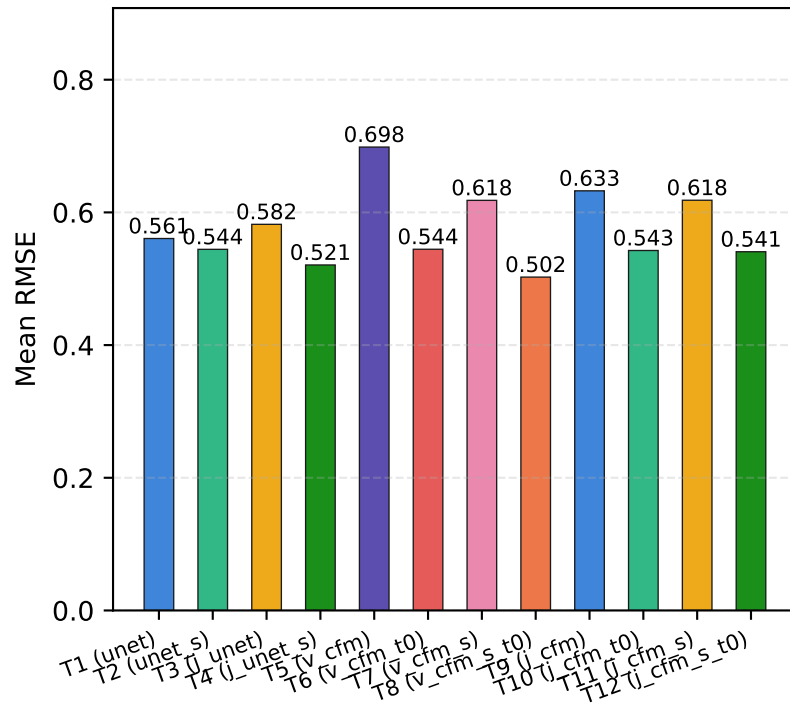
ID	sigma	rho	beta	c1
T3_joint_direct_unet_s0s1	0.1388	0.2611	0.0914	0.0144
T4_joint_direct_unet_s0s1_small	0.1520	0.4437	0.1059	0.0099
T9_joint_cfm_s0s1	0.1055	0.1703	0.0750	0.0214
T10_joint_cfm_s0s1_tau0	7.6847	28.1945	1.7799	0.5926
T11_joint_cfm_s0s1_small	0.1233	0.2539	0.0716	0.0151
T12_joint_cfm_s0s1_small_tau0	3.1293	27.6088	2.8233	0.2134

--- S1 ---

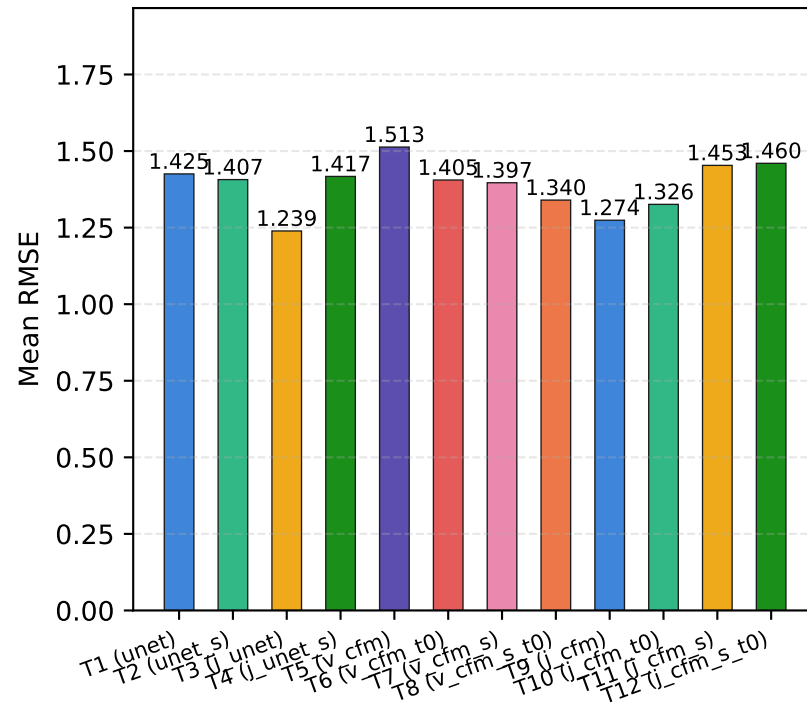
ID	sigma	rho	beta	c1
T3_joint_direct_unet_s0s1	1.2000	3.3023	0.5006	0.0106
T4_joint_direct_unet_s0s1_small	1.3839	3.9869	0.3948	0.0179
T9_joint_cfm_s0s1	1.2535	3.5314	0.3279	0.0161
T10_joint_cfm_s0s1_tau0	7.6184	28.0280	1.6642	0.6060
T11_joint_cfm_s0s1_small	1.2742	3.6251	0.3757	0.0131
T12_joint_cfm_s0s1_small_tau0	4.7978	27.8122	3.3091	0.2152

S0 / S1 Mean RMSE & Robustness

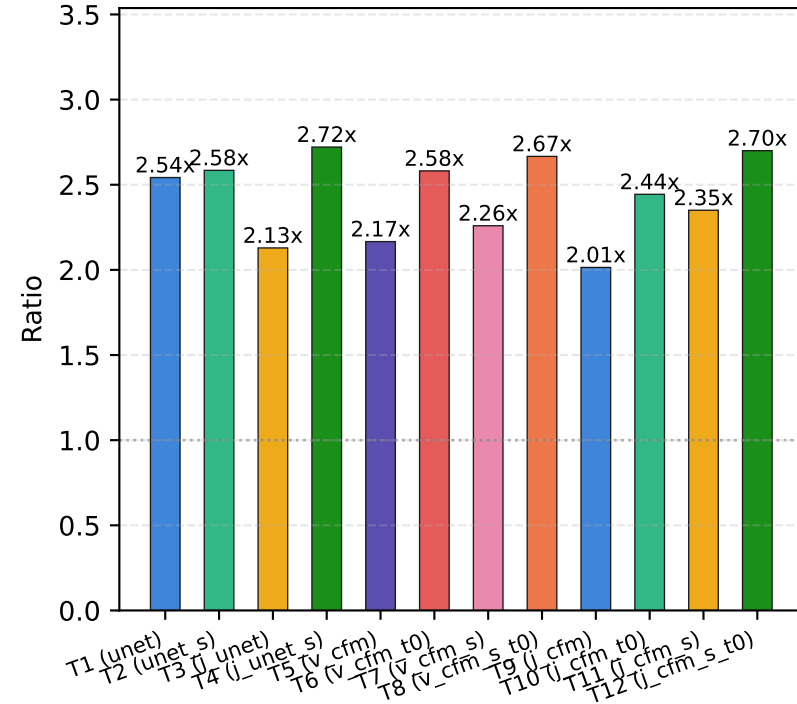
S0 Mean RMSE



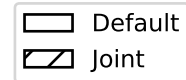
S1 Mean RMSE



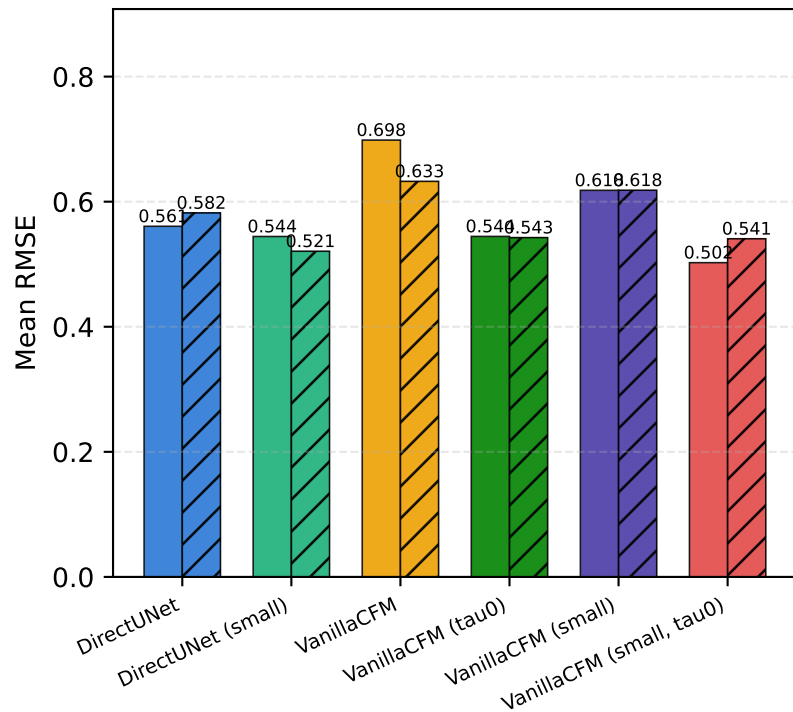
Degradation (S1/S0)



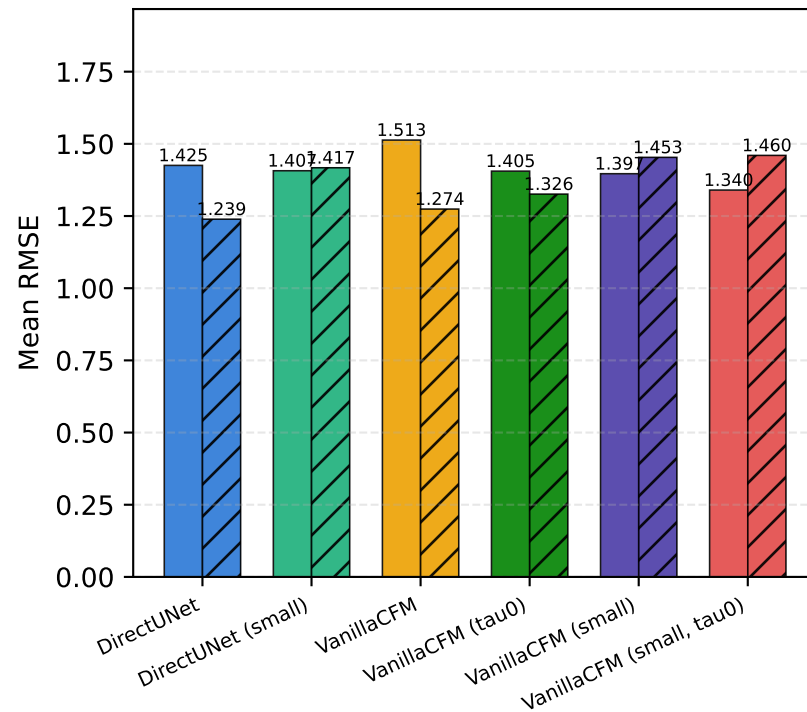
S0 / S1 Mean RMSE & Robustness — Default vs. Joint



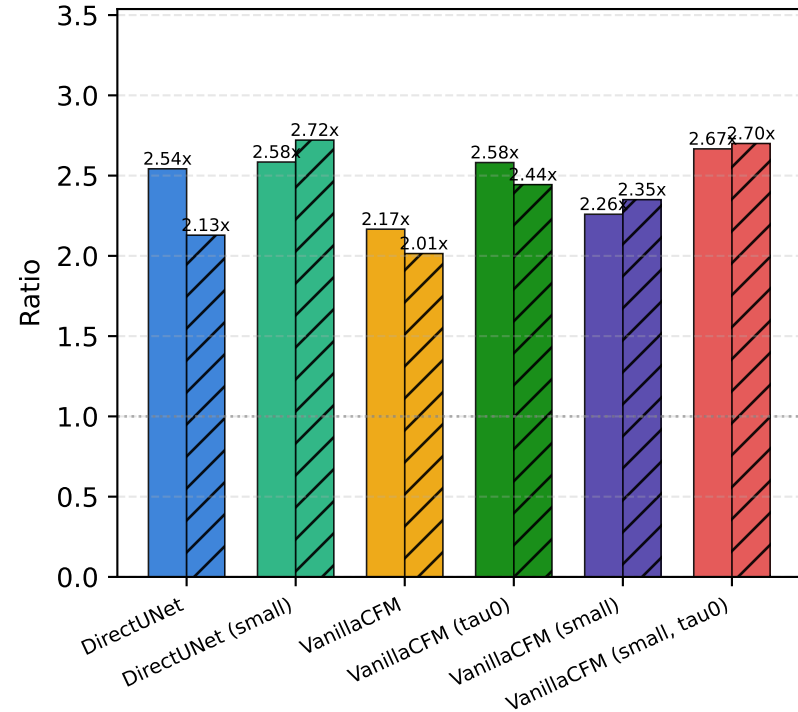
S0 Mean RMSE



S1 Mean RMSE

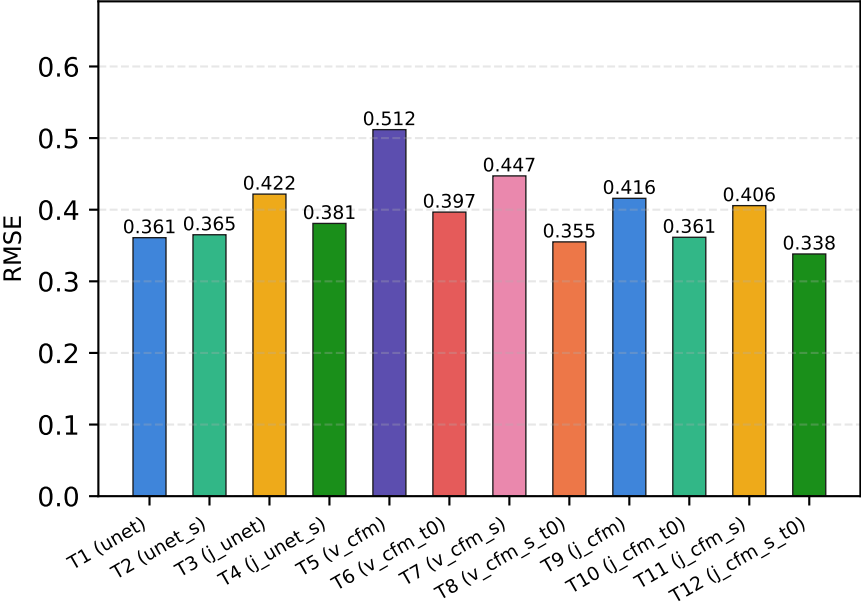


Degradation (S1/S0)

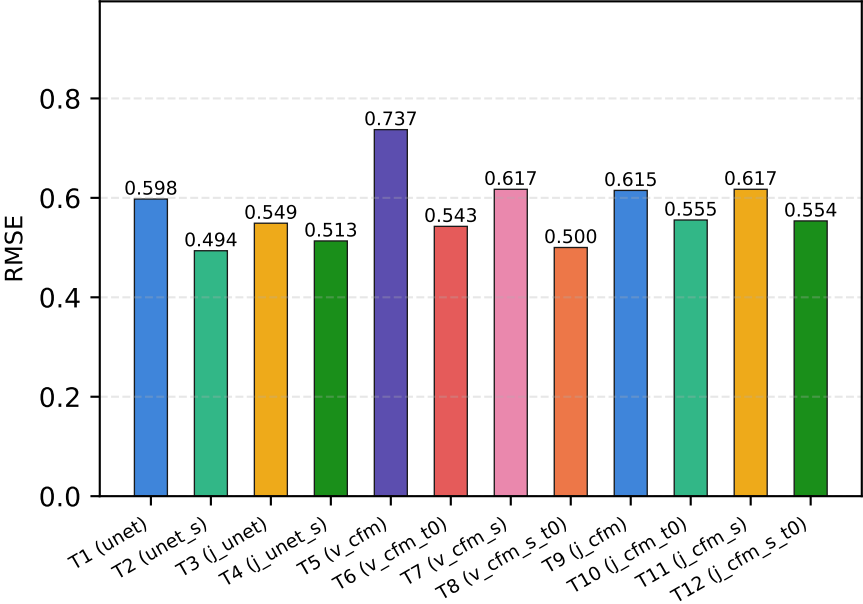


Per-Component RMSE Breakdown (S0, S1)

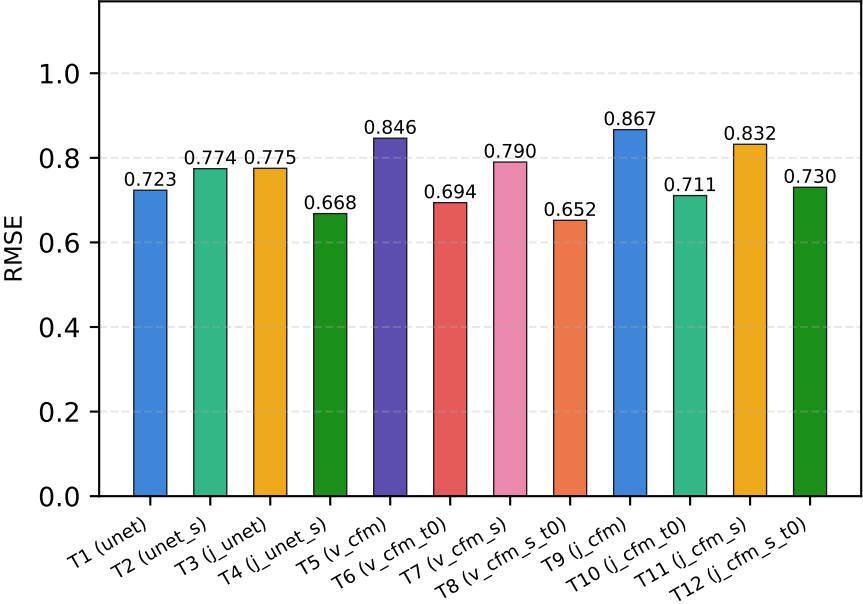
S0 — X



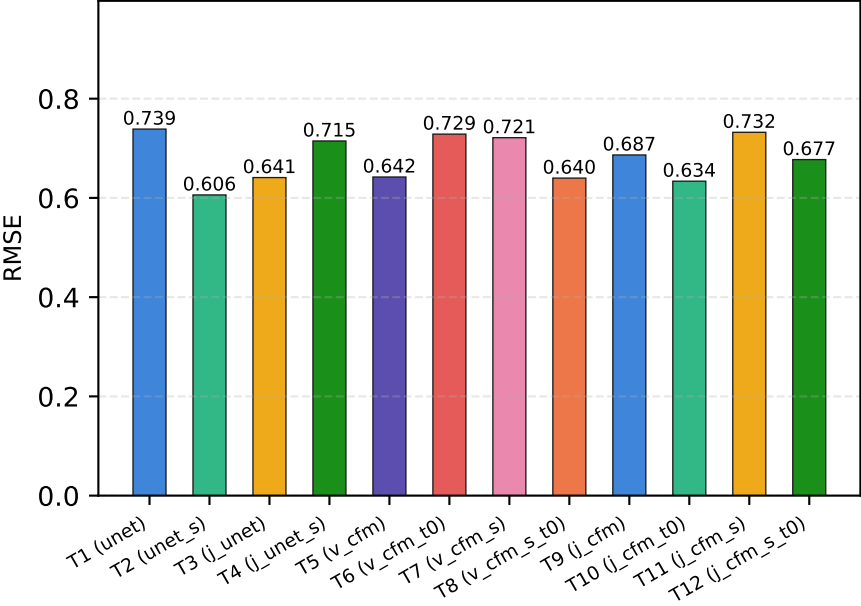
S0 — Y



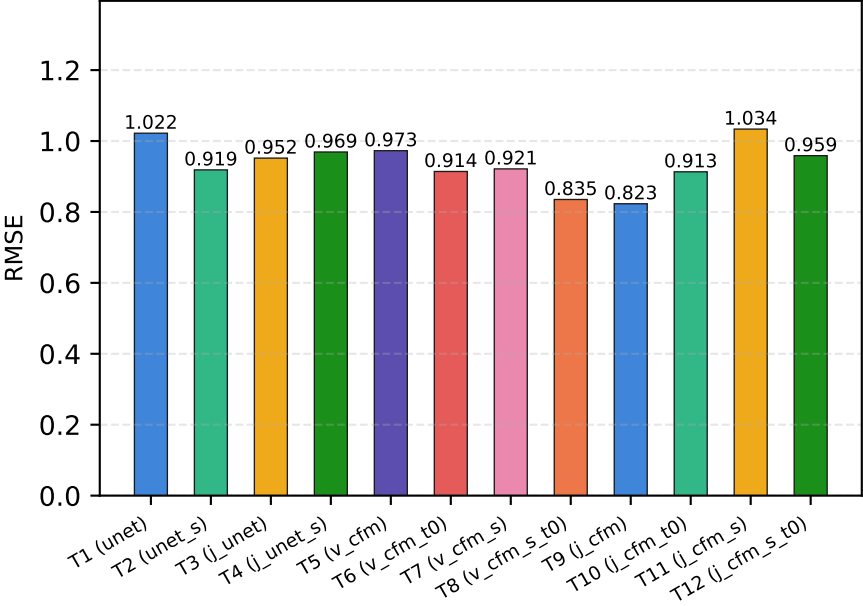
S0 — Z



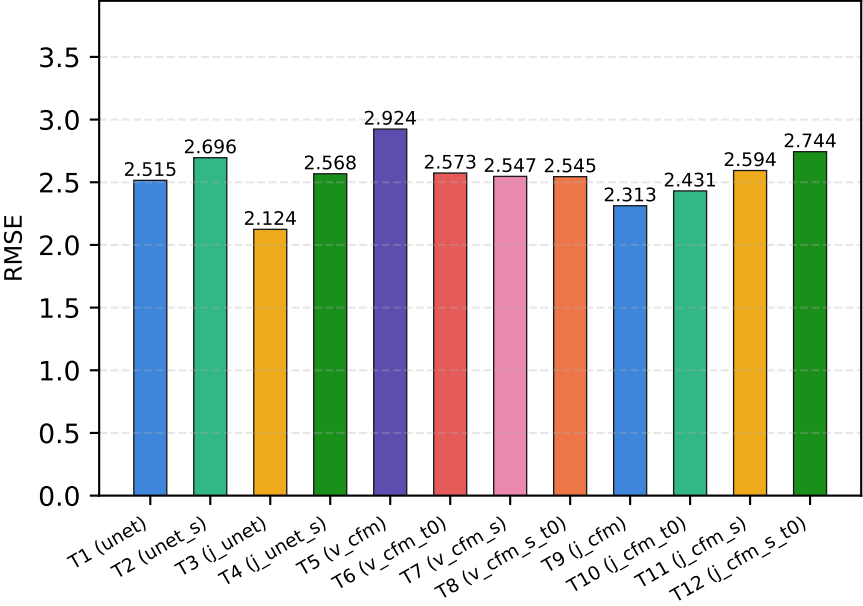
S1 — X



S1 — Y

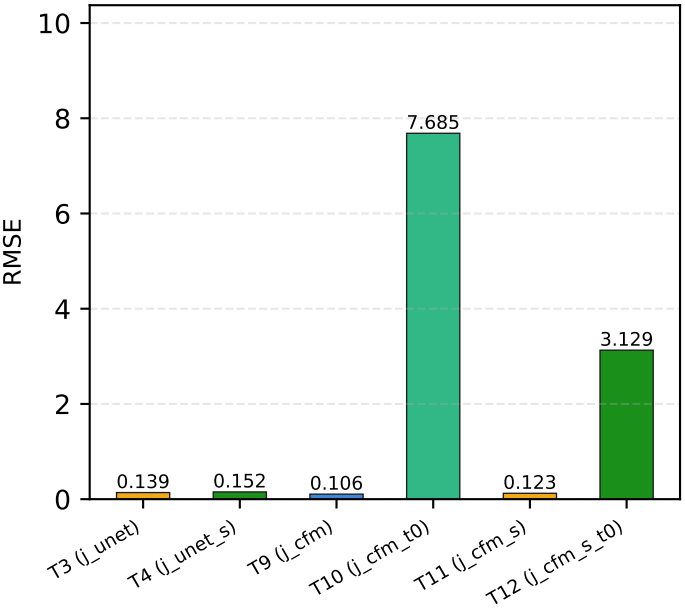


S1 — Z

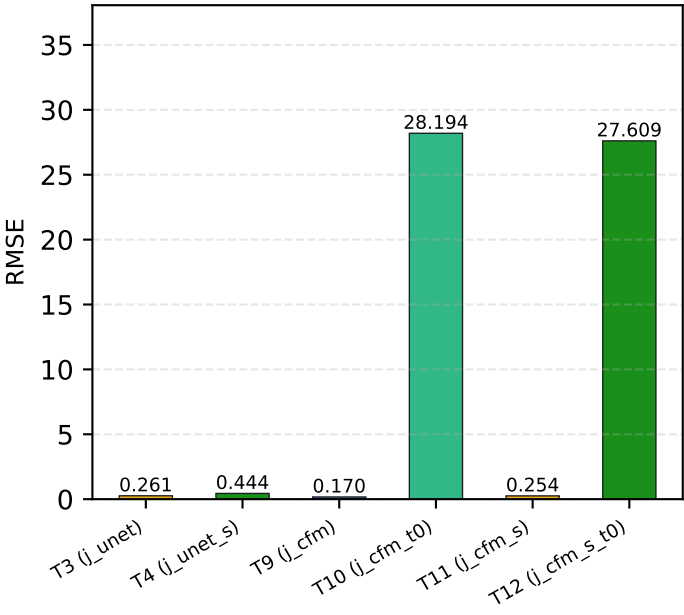


Parameter RMSE Breakdown (S0, S1) — Joint Variants

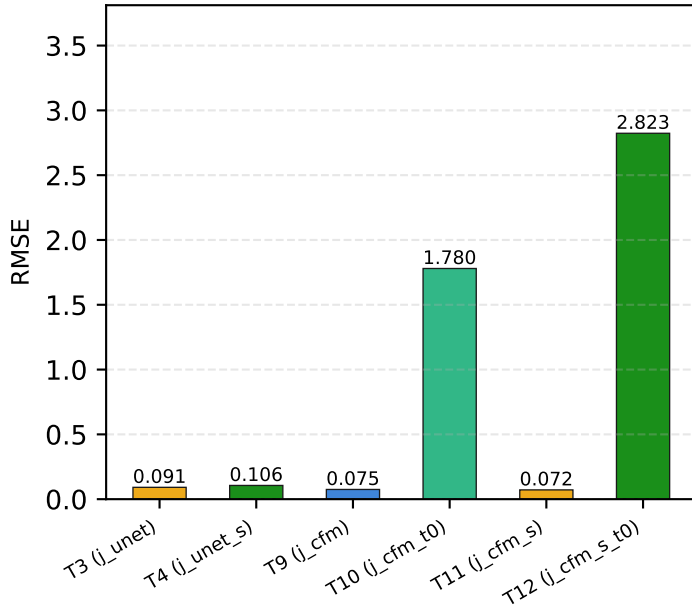
S0 — sigma



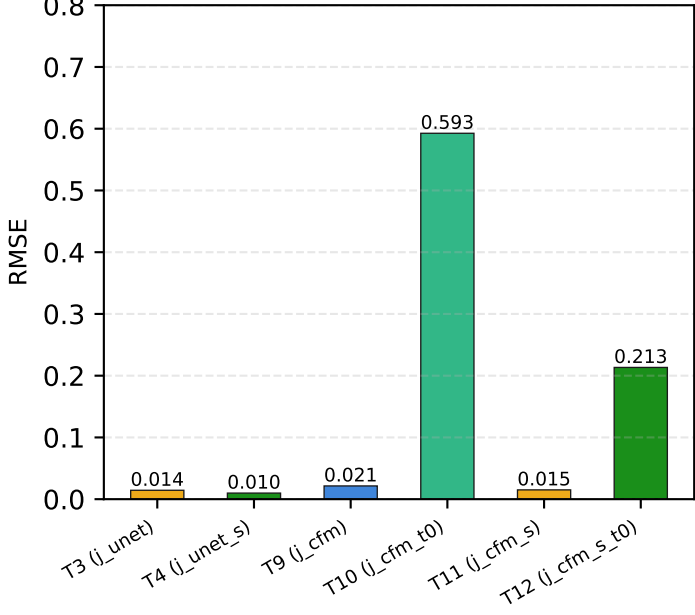
S0 — rho



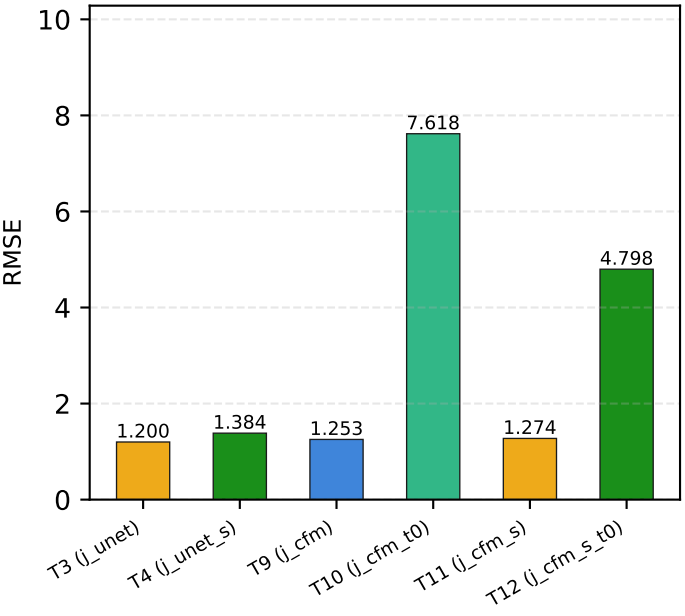
S0 — beta



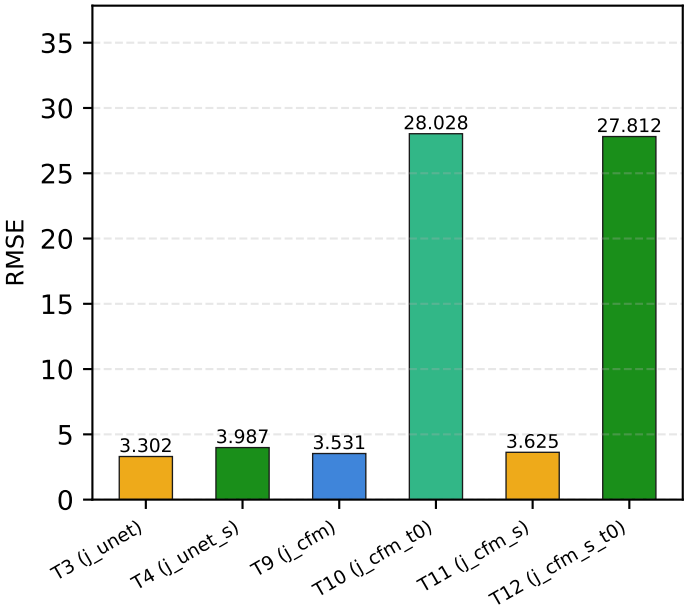
S0 — c1



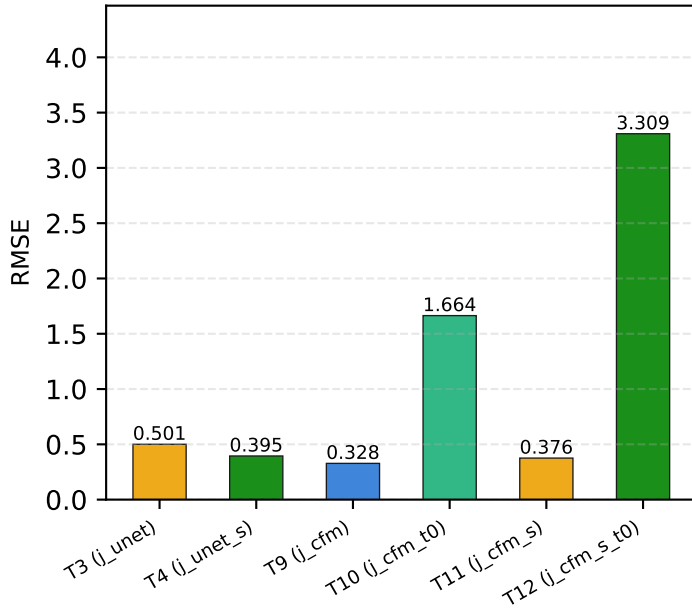
S1 — sigma



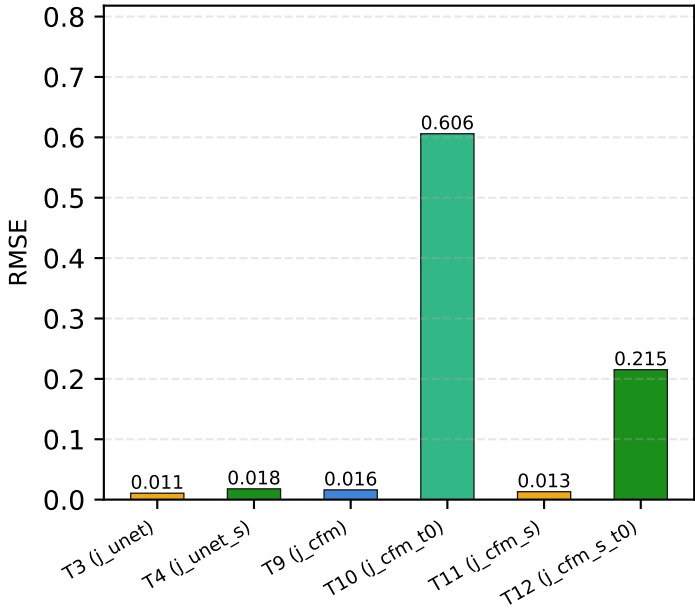
S1 — rho



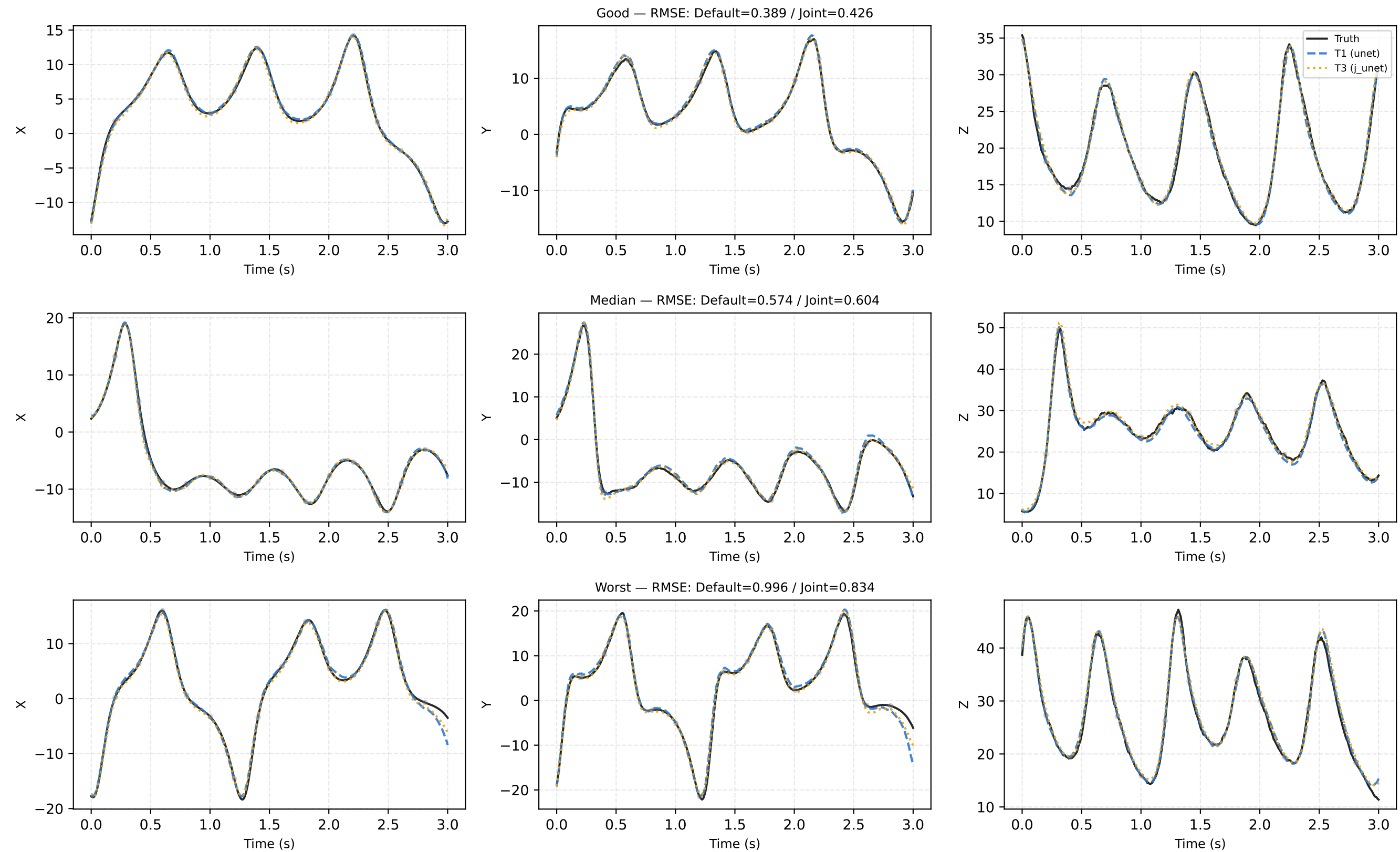
S1 — beta



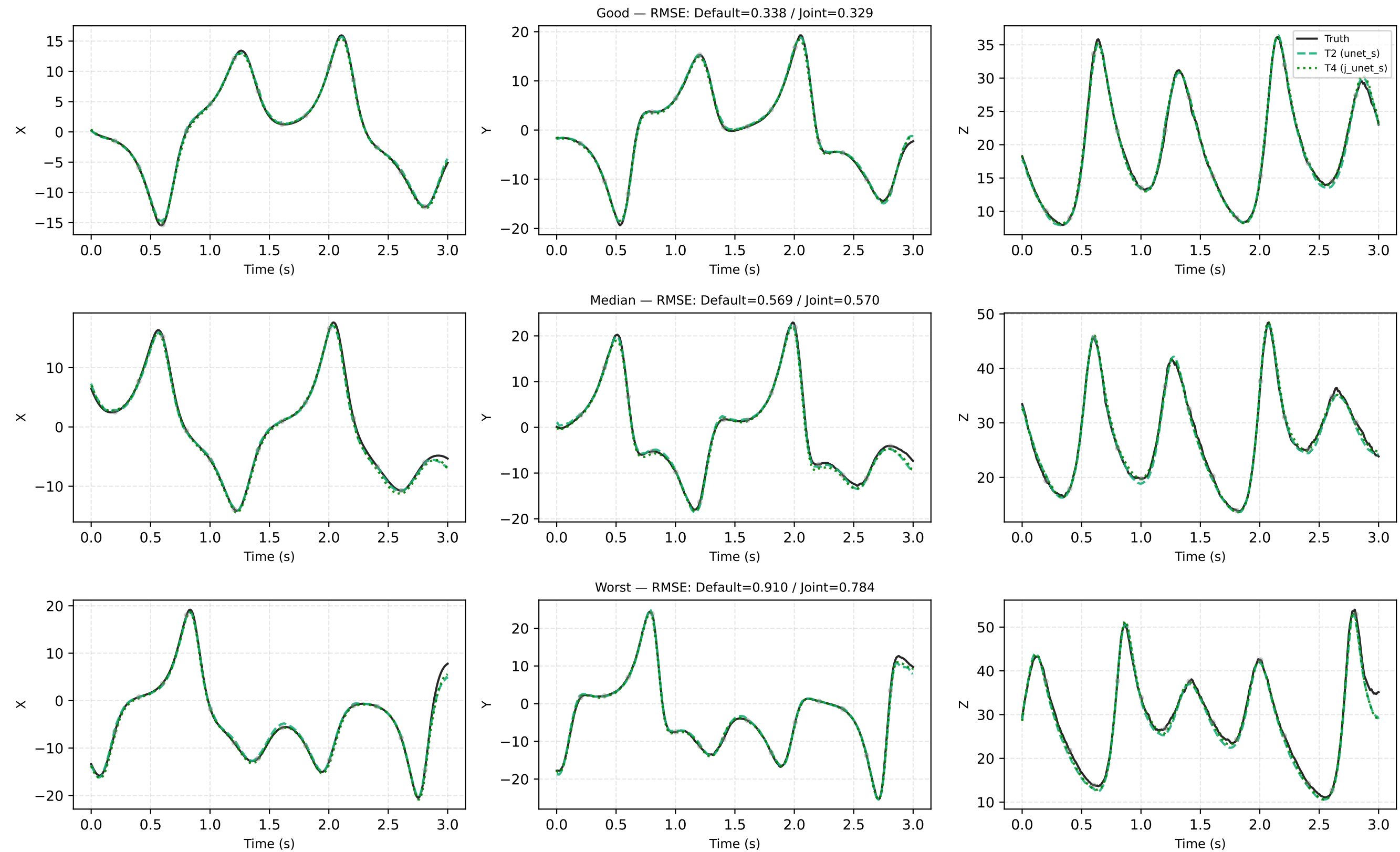
S1 — c1



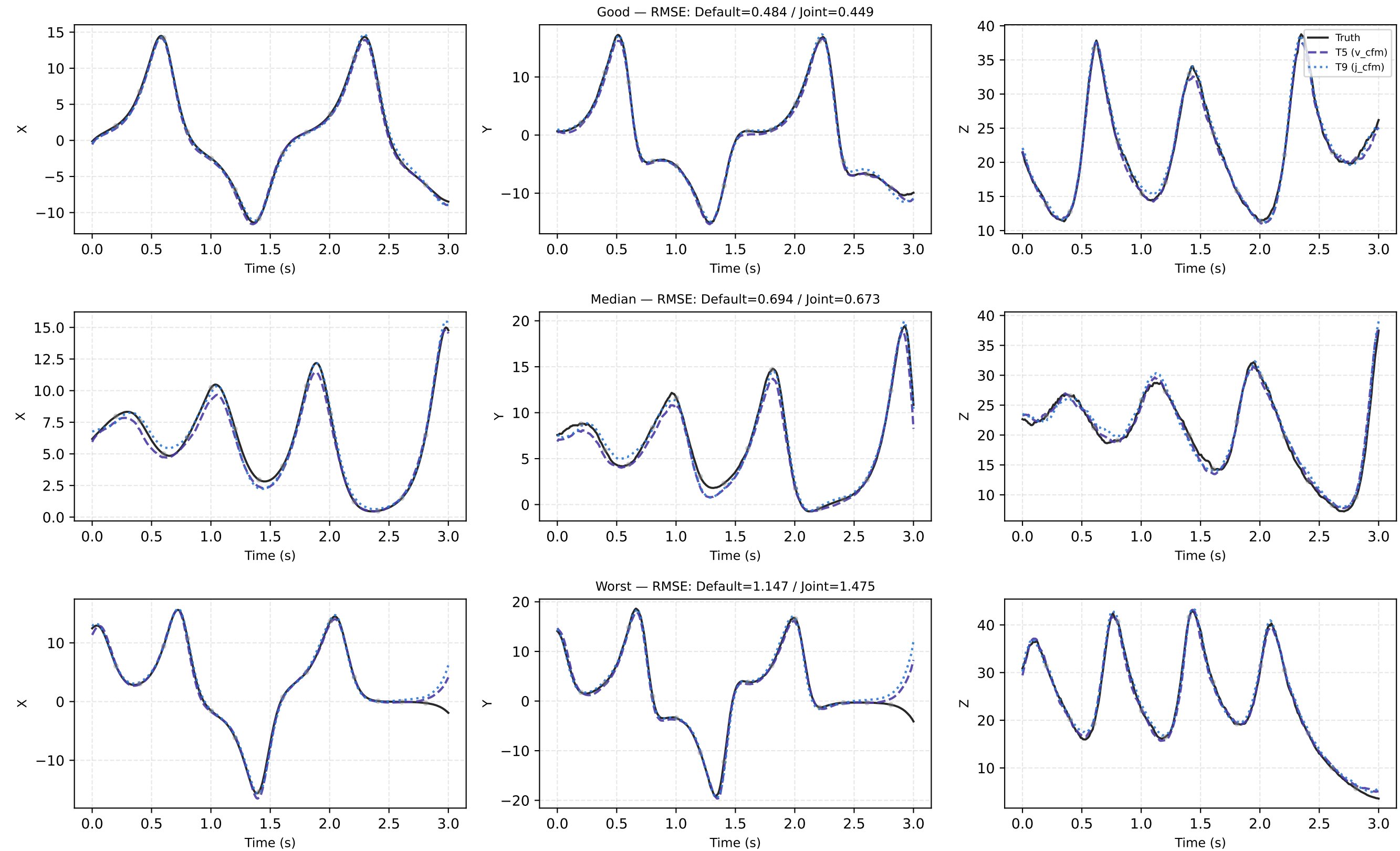
S0 — DirectUNet (default): Default vs. Joint (good / median / worst reconstructions)



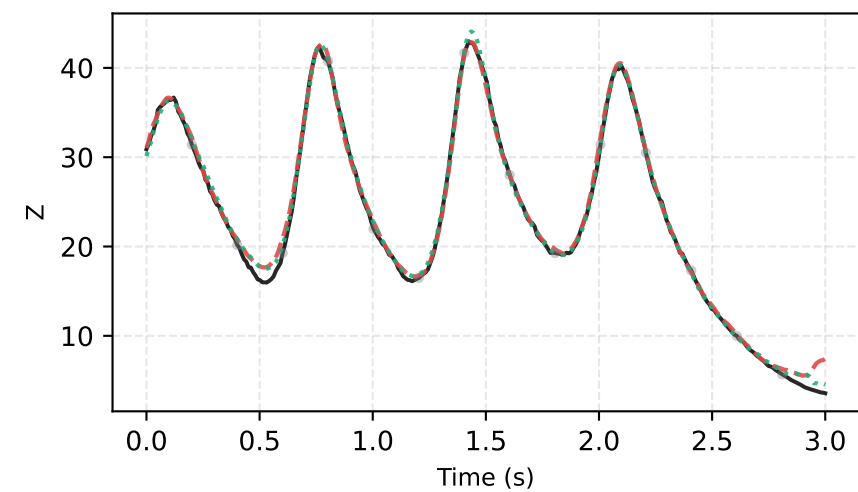
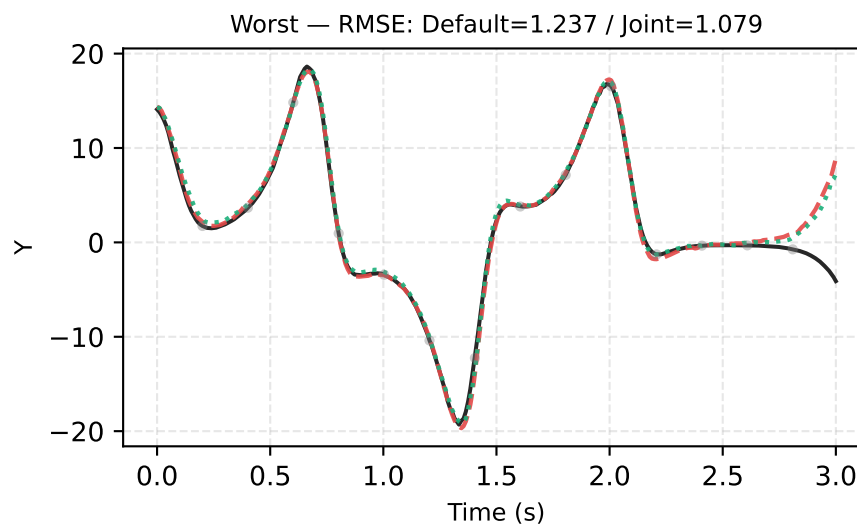
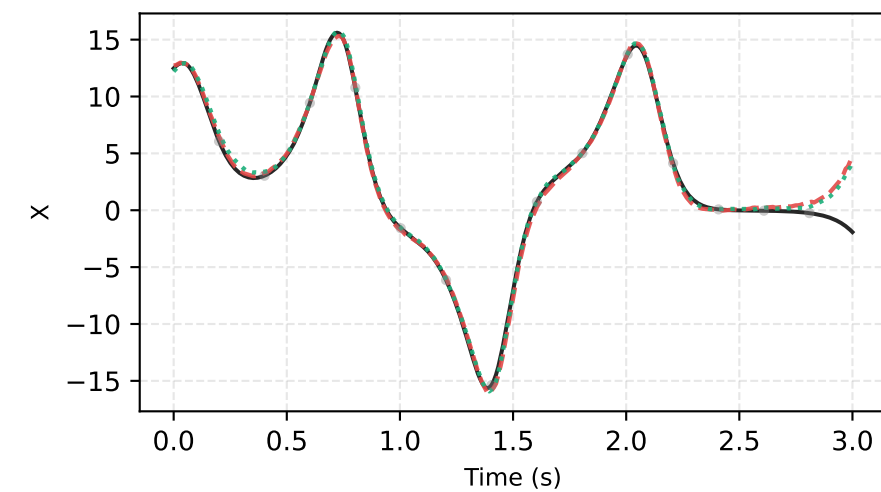
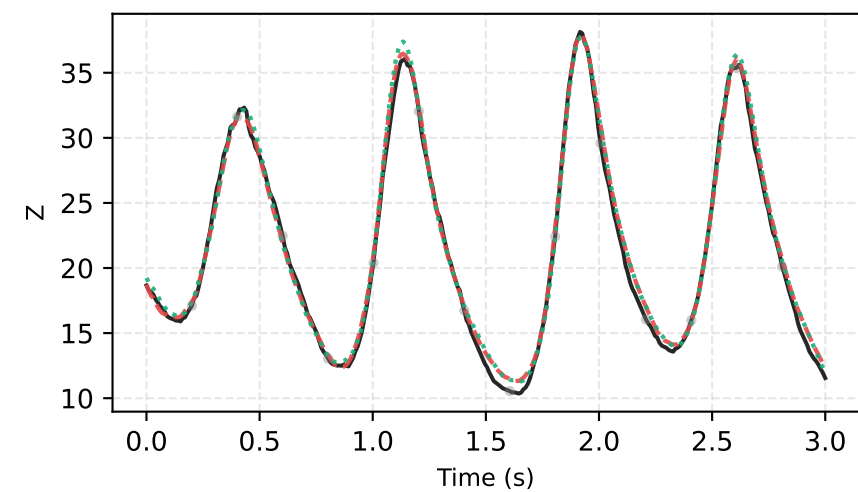
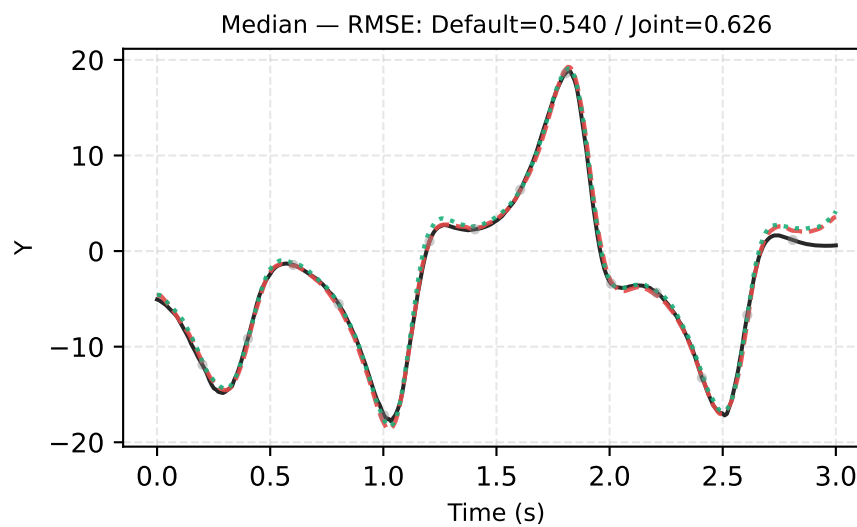
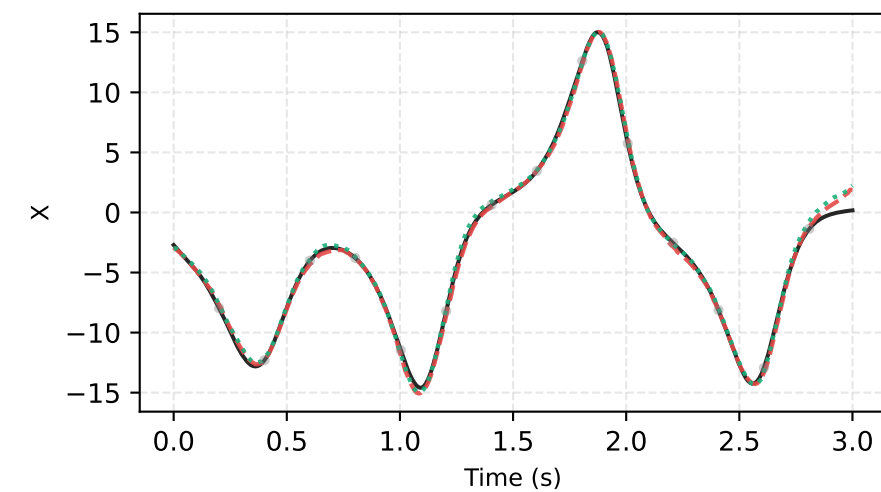
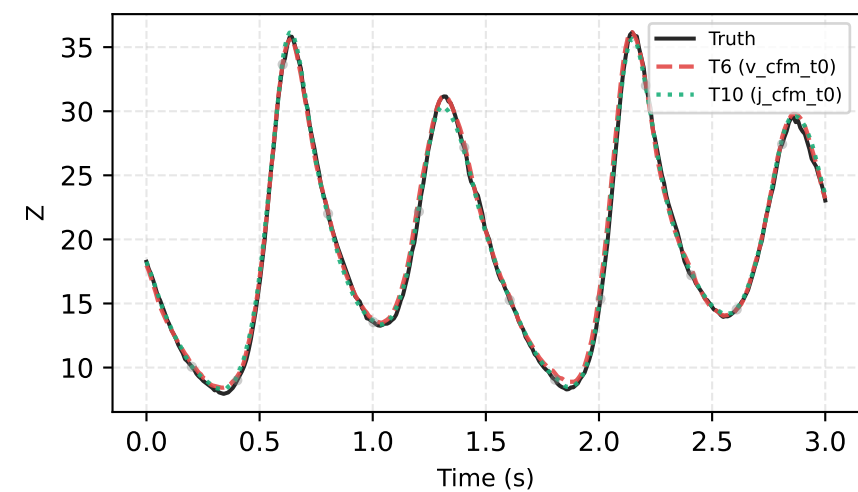
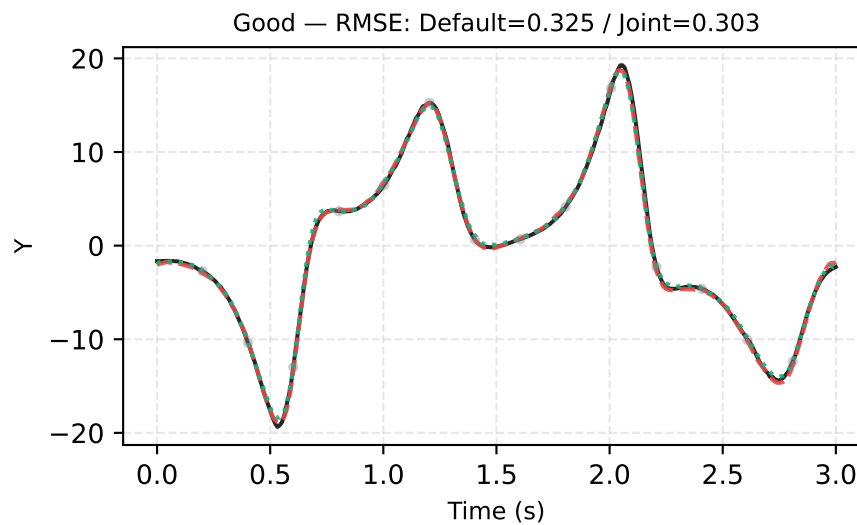
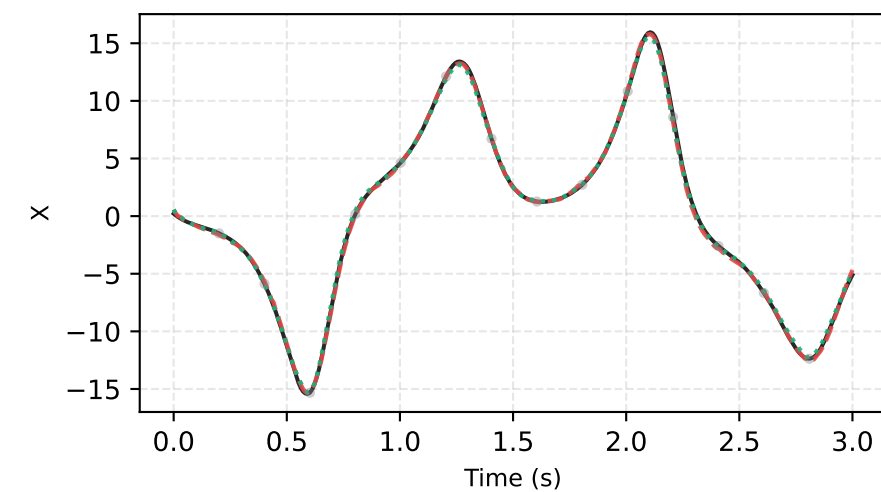
S0 — DirectUNet (small): Default vs. Joint (good / median / worst reconstructions)



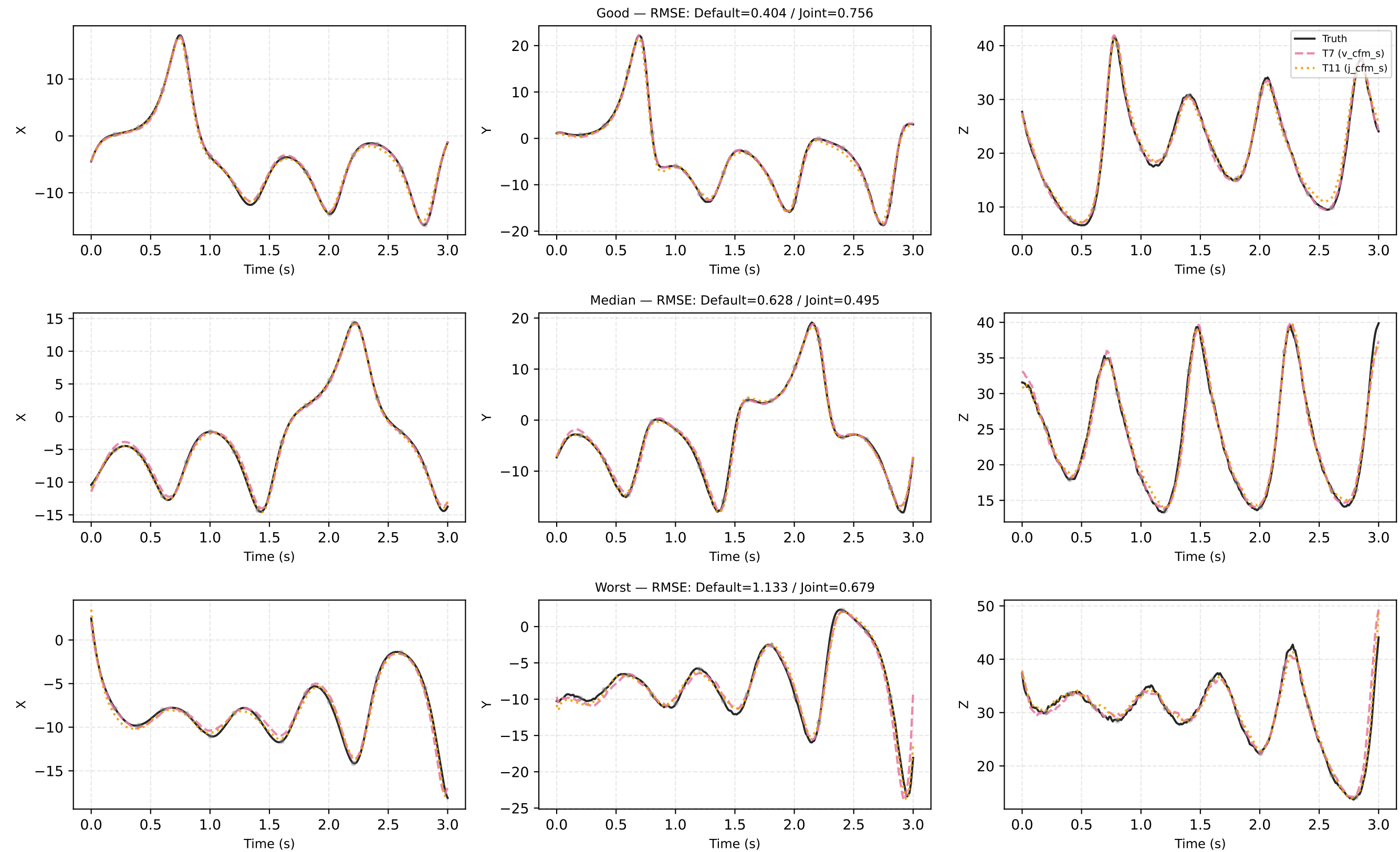
S0 — VanillaCFM (default): Default vs. Joint (good / median / worst reconstructions)



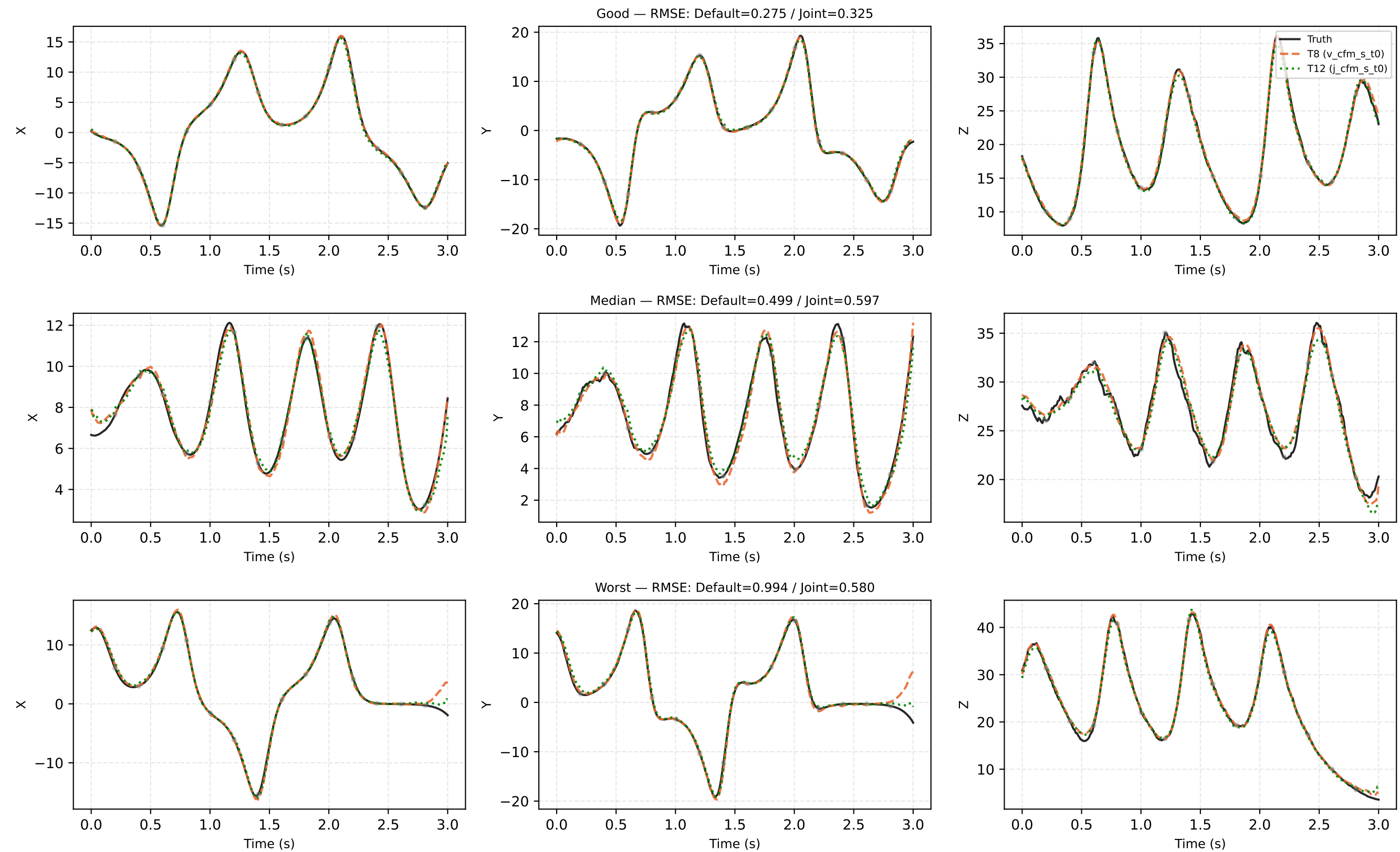
S0 — VanillaCFM (default, tau0): Default vs. Joint (good / median / worst reconstructions)



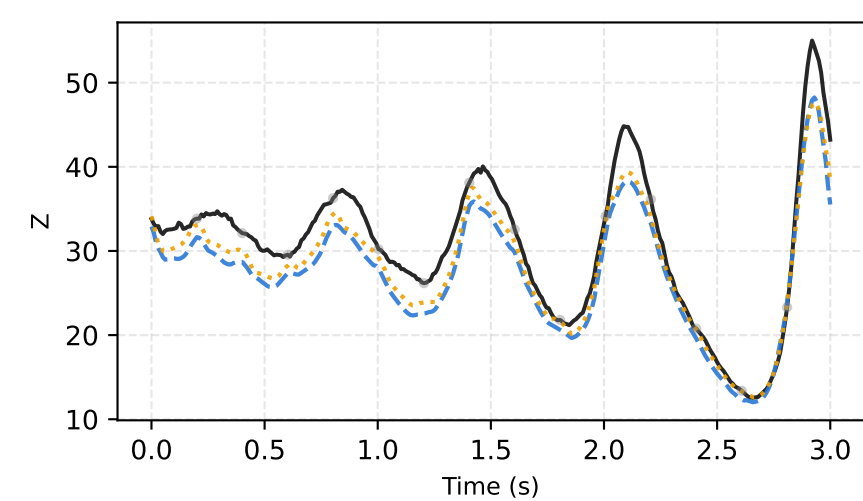
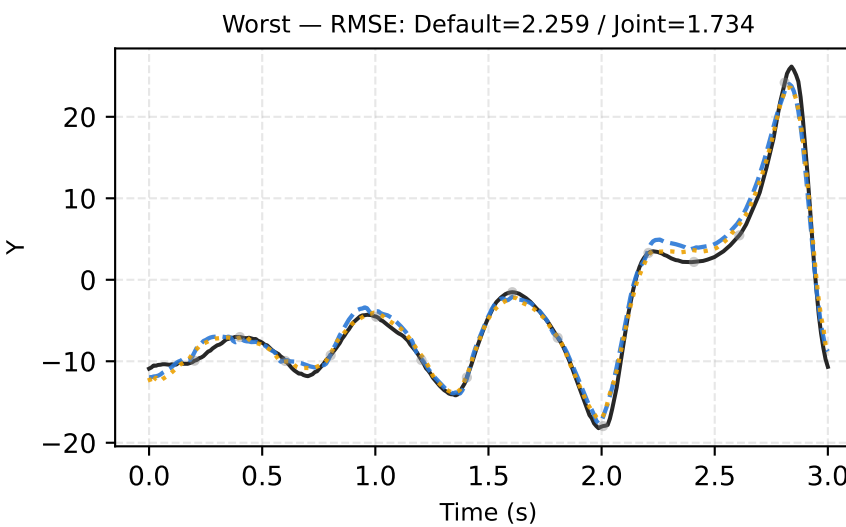
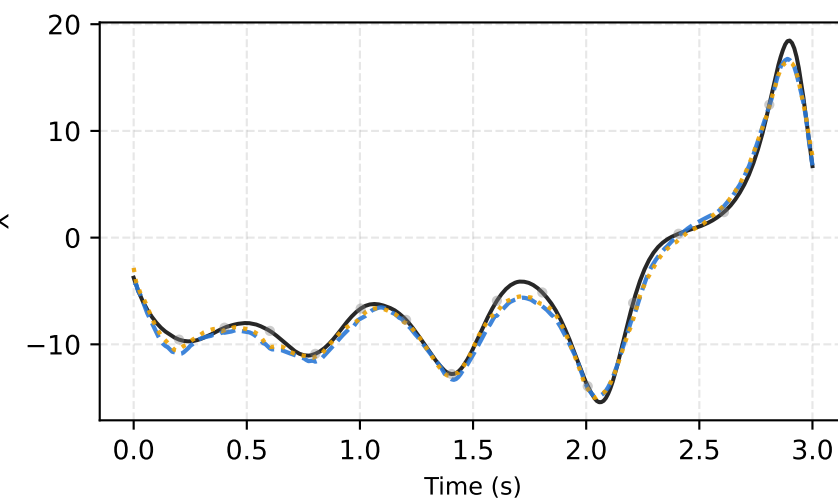
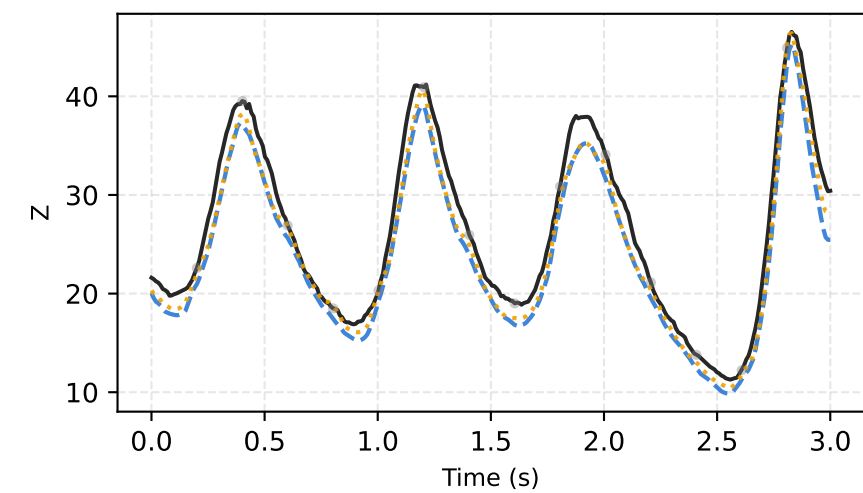
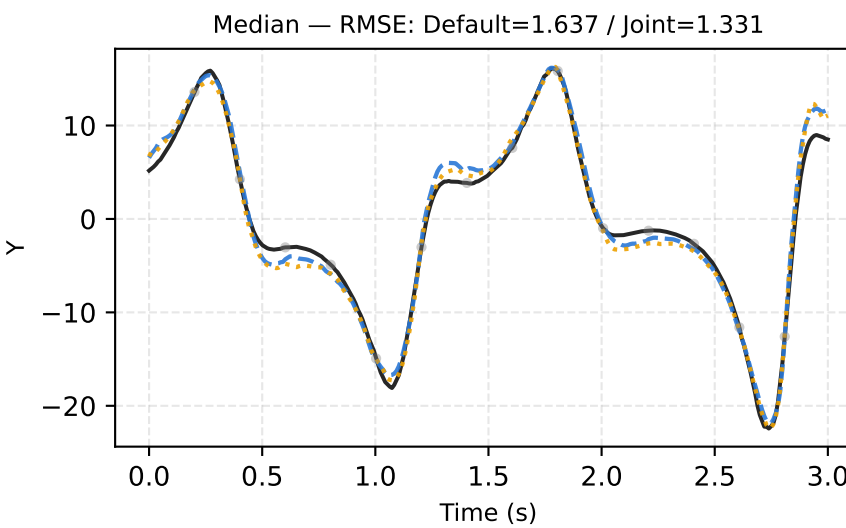
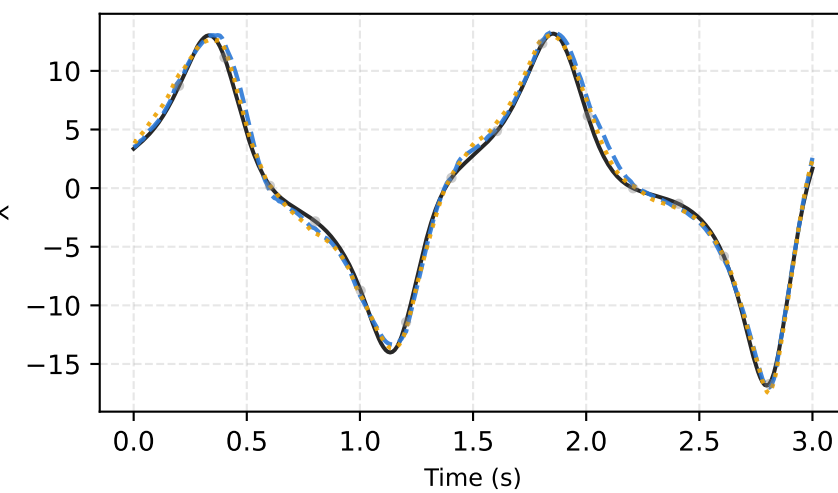
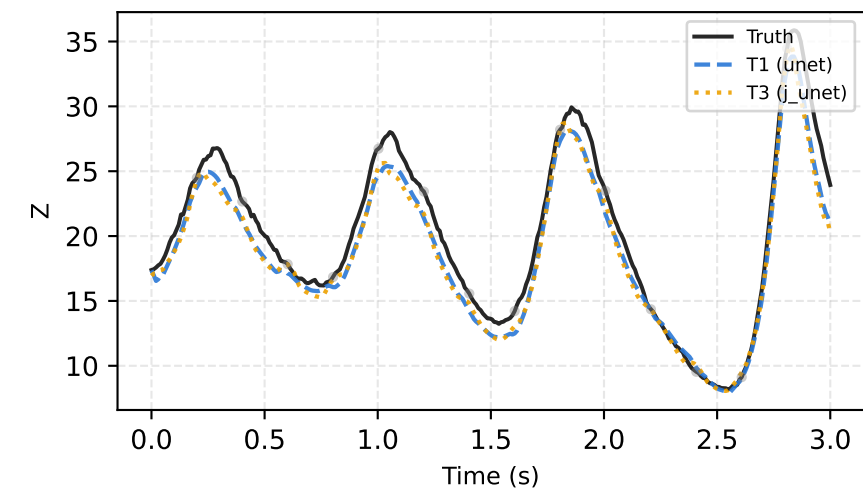
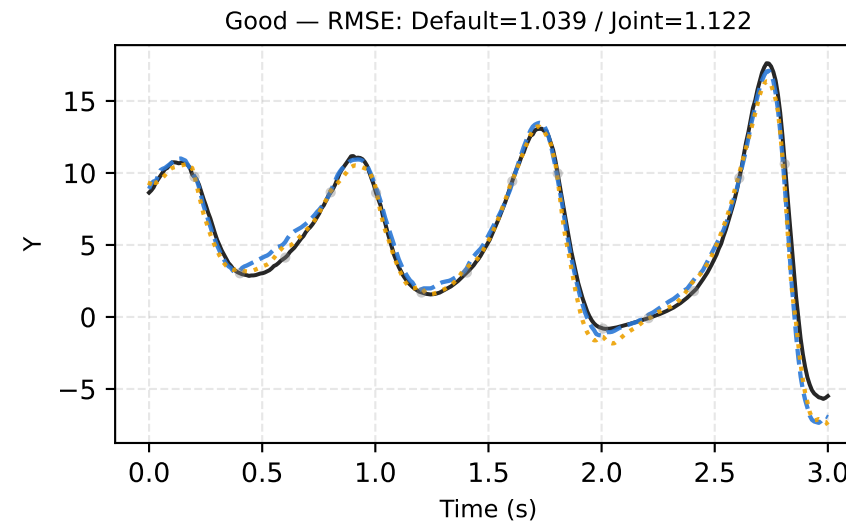
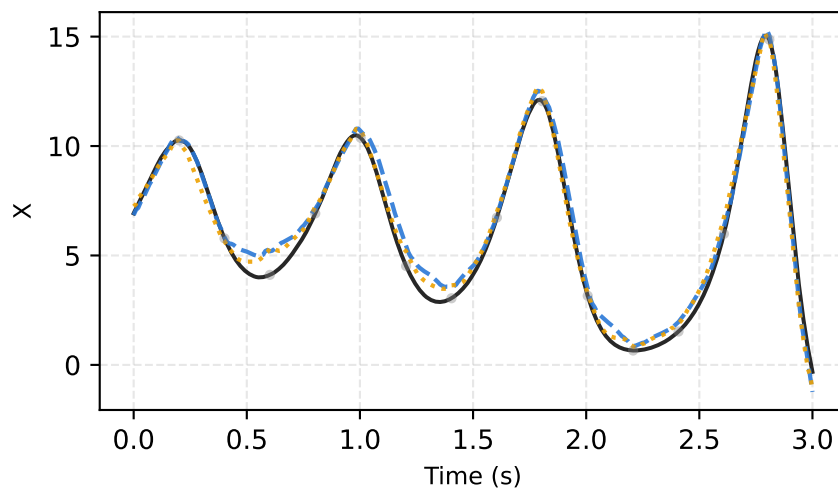
S0 — VanillaCFM (small): Default vs. Joint (good / median / worst reconstructions)



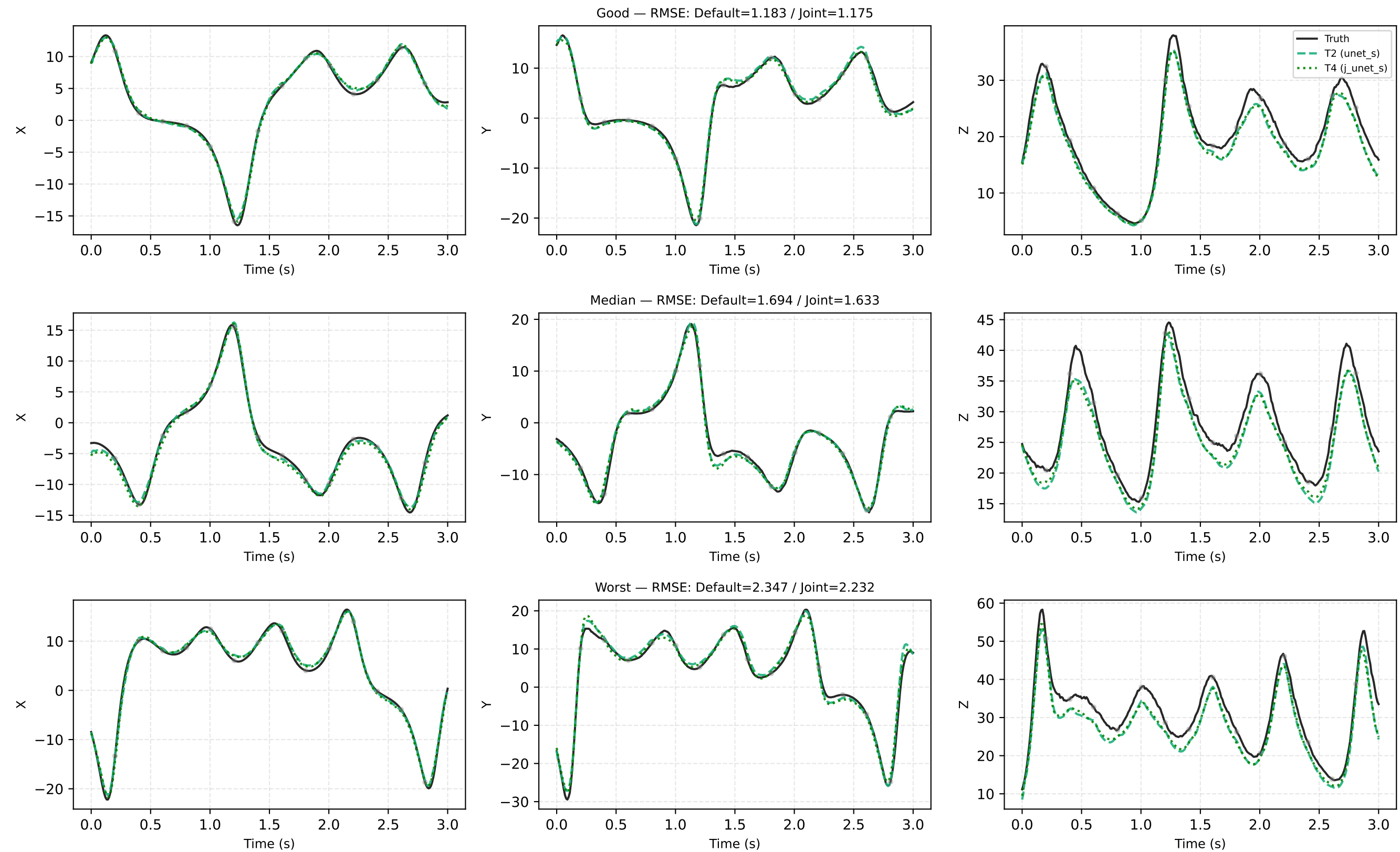
S0 — VanillaCFM (small, tau0): Default vs. Joint (good / median / worst reconstructions)



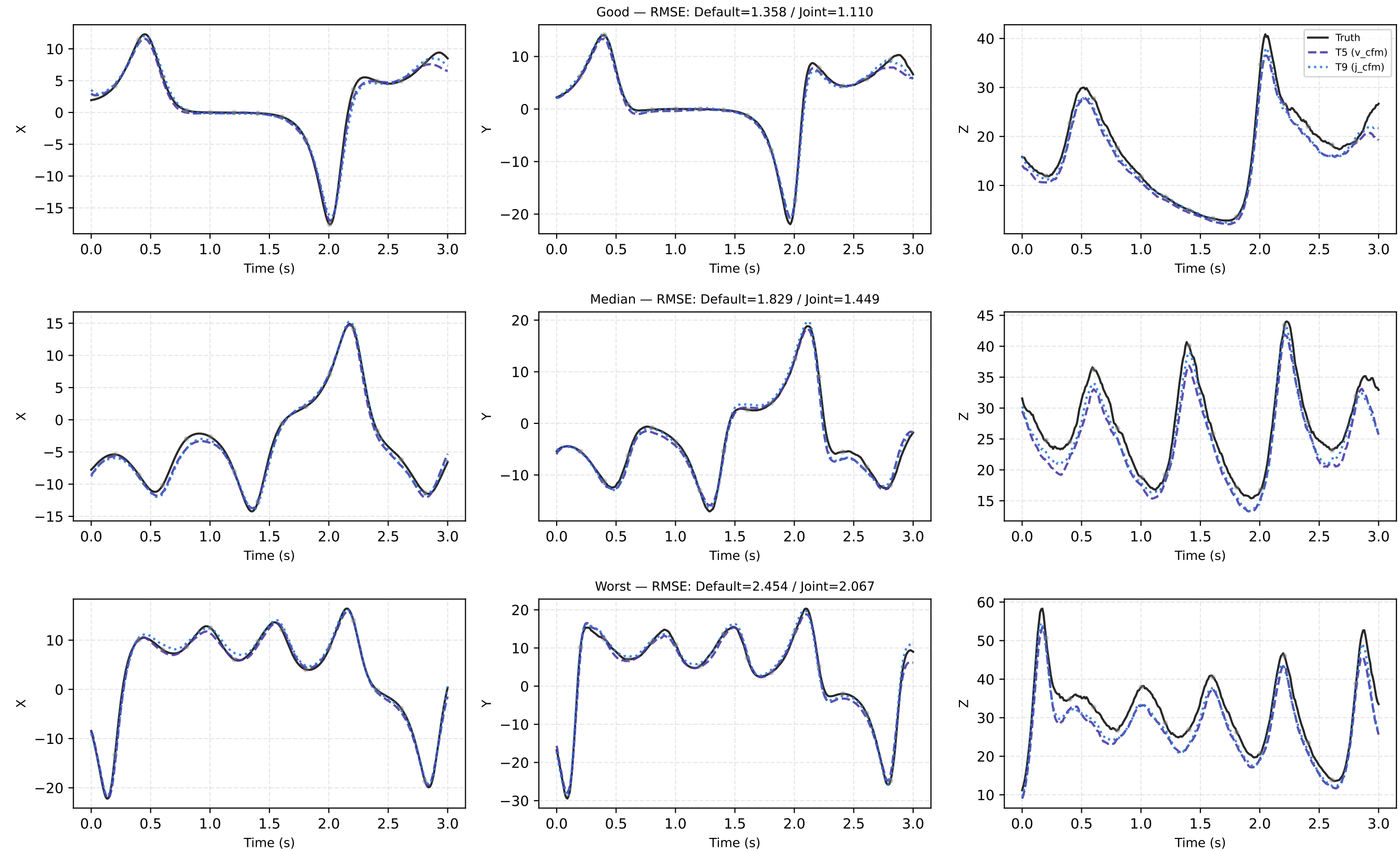
S1 — DirectUNet (default): Default vs. Joint (good / median / worst reconstructions)



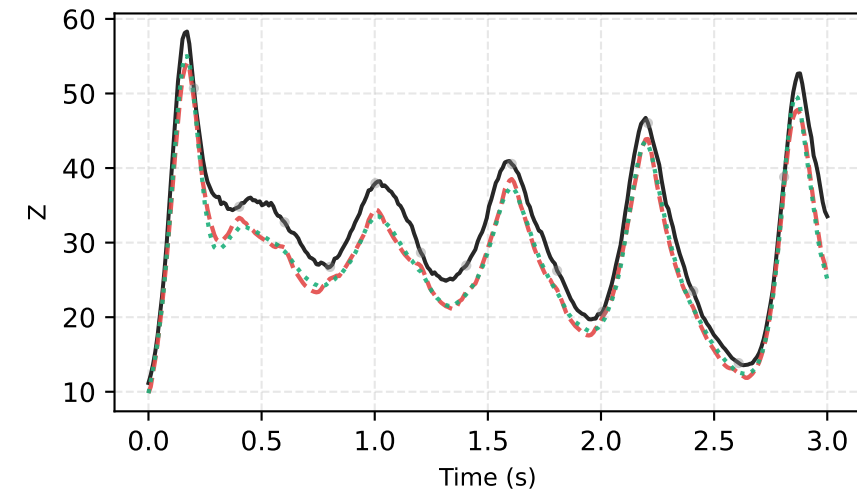
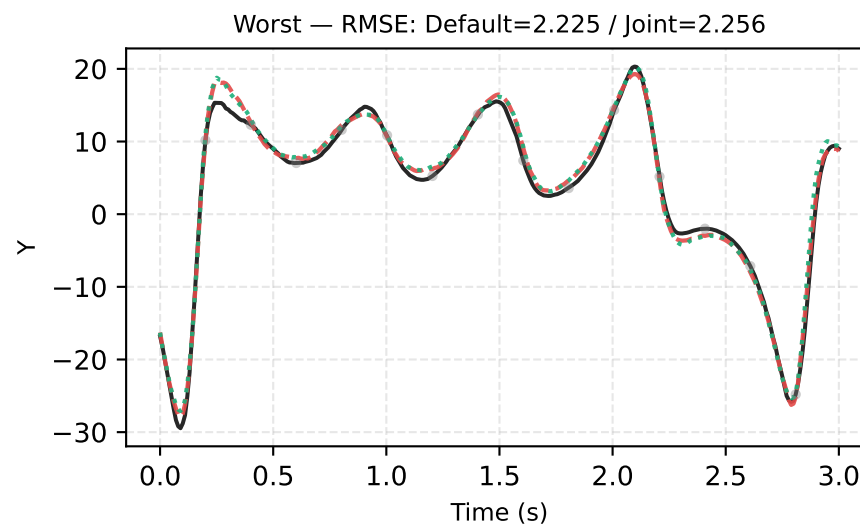
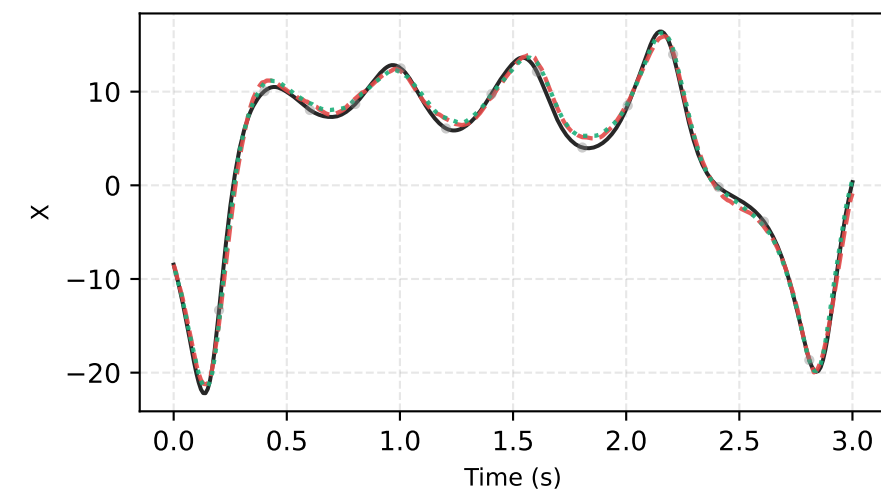
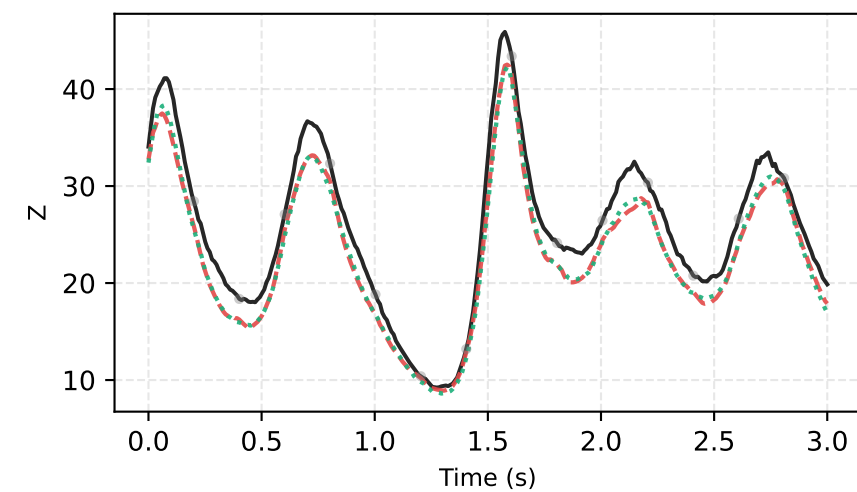
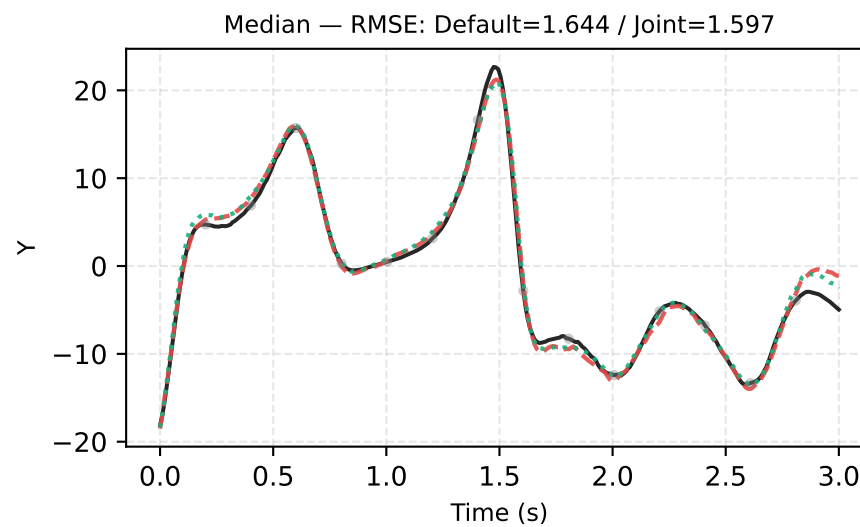
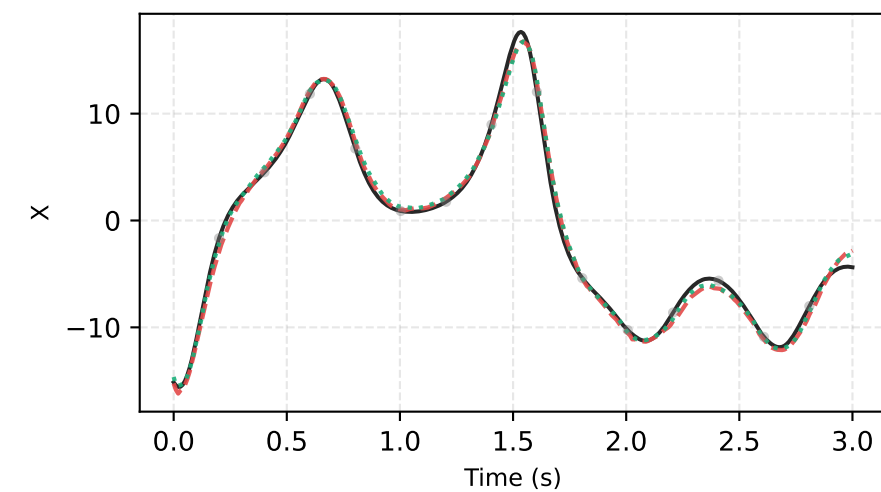
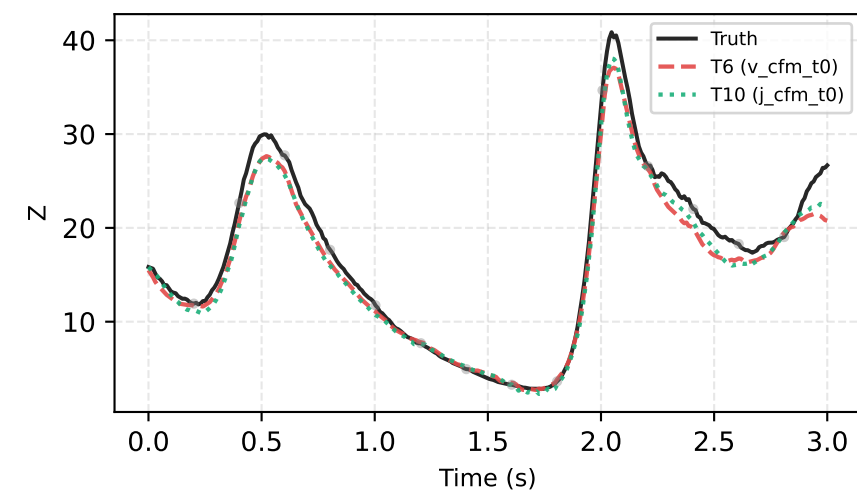
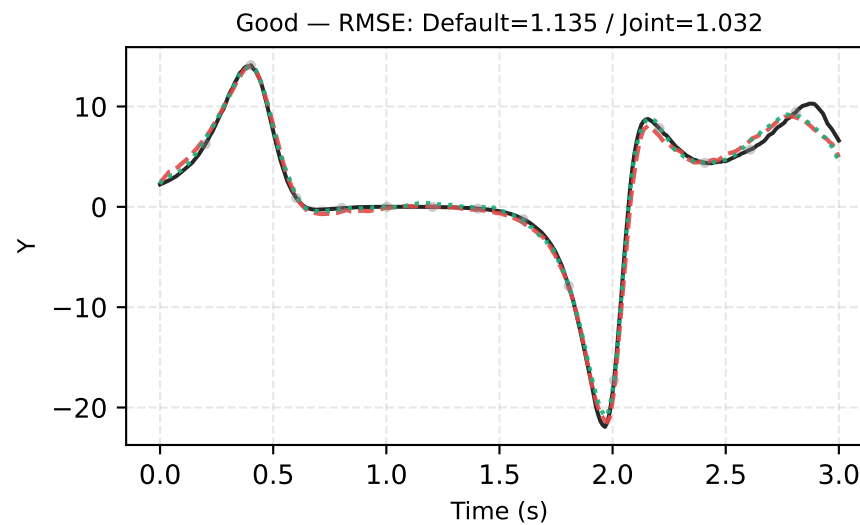
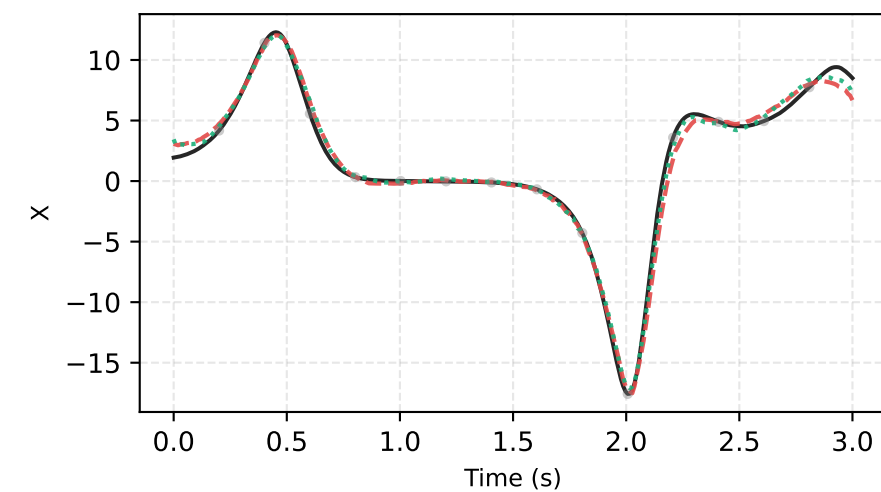
S1 — DirectUNet (small): Default vs. Joint (good / median / worst reconstructions)



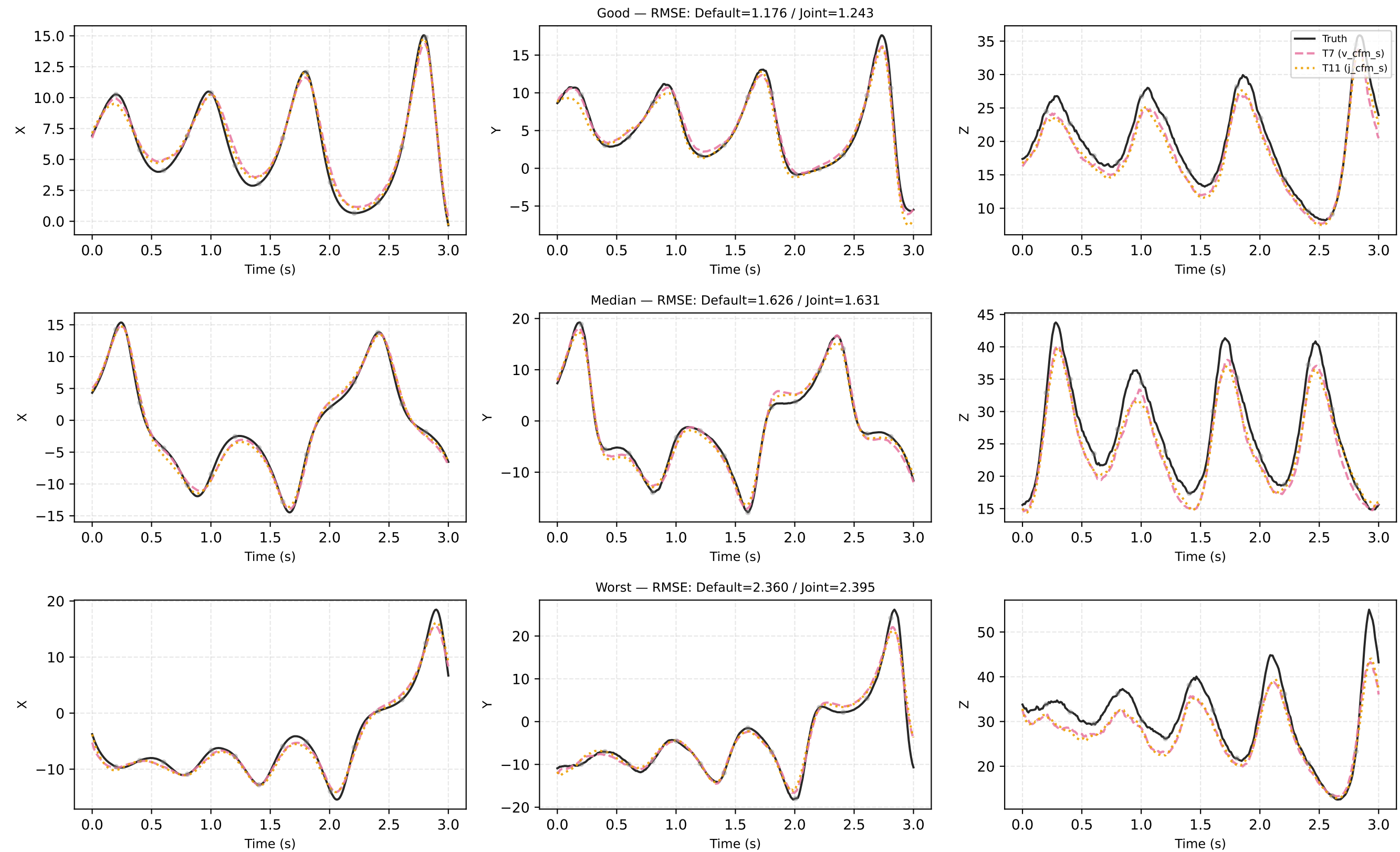
S1 — VanillaCFM (default): Default vs. Joint (good / median / worst reconstructions)



S1 — VanillaCFM (default, tau0): Default vs. Joint (good / median / worst reconstructions)



S1 — VanillaCFM (small): Default vs. Joint (good / median / worst reconstructions)



S1 — VanillaCFM (small, tau0): Default vs. Joint (good / median / worst reconstructions)

